

Horizon Entropy and Gravitational Decoherence: $\mu < 1, \Sigma = 1$ a Dual-Sector Cosmology 0 Parameters Beyond Λ CDM with Full Planck 2018 Validation

The Informational Actualization Model

Heath W. Mahaffey

Independent Researcher

Entiat, Washington, USA

hmahaffeyges@gmail.com

<https://github.com/hmahaffeyges/IAM-Validation>

March 2026

Abstract

Jacobson showed in 1995 that Einstein's field equations are an equation of state derived from $\delta Q = T dS$ applied to local causal horizons, with entropy proportional to horizon area. Cai and Kim extended this in 2005 to the FRW apparent horizon, recovering the Friedmann equations by the same thermodynamic argument. Both derivations assume the entropy functional contains only geometric terms. We identify a specific additional contribution that has been present but unaccounted for throughout the history of the universe: the irreversible classical information produced by gravitational decoherence of matter on timelike worldlines. This information carries a minimum thermodynamic cost set by Landauer's principle and is encoded on the cosmic horizon by the holographic principle. Including it in the Jacobson–Cai–Kim entropy functional yields, with no additional free parameters, a dual-sector gravitational coupling: $\mu(a) < 1$ for matter perturbations and $\Sigma(a) = 1$ for photon propagation. The sector split is not imposed; it follows causally from the metric signature, since decoherence requires nonzero proper time and photons propagate on null worldlines with $\Delta\tau = 0$. The coupling constant $\beta_m = \Omega_m/2$ is derived from the virial theorem and consistent with six independent published N-body simulation studies to within 1%. The activation function $E(a) = \exp(1 - 1/a)$ is derived from the ratio of structure-formation rate to cosmic horizon area and recovered to within 1% by the Sheth–Tormen halo mass function at galaxy-scale halos. The resulting perturbation-level modification to Λ CDM has been validated against the full Planck 2018 likelihood through 17 converged MCMC chains (Gelman–Rubin $R - 1 < 0.01$ throughout), yielding $\Delta\chi^2 = +0.54$ best-fit relative to Λ CDM under the Planck CamSpec TTTEEE likelihood, with σ_8 shifting from 0.809 to 0.800 and an inferred matter-sector Hubble constant $H_0^{\text{matter}} = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$ alongside $H_0^{\text{photon}} = 67.16 \text{ km s}^{-1} \text{ Mpc}^{-1}$, both within 1σ of their respective observational anchors. All standard cosmological parameters shift by less than 0.1σ . The model introduces zero free parameters beyond Λ CDM. Specific falsifiable predictions for Euclid, DESI Year 5, and CMB-S4 are stated. All chains, modified Boltzmann code, configuration files, and analysis scripts are publicly archived at the GitHub repository listed above.

Contents

1	Introduction	4
2	The Physical Mechanism	4
2.1	Why Photons Are Exempt	5
3	Thermodynamic Derivation	6
3.1	Jacobson’s Derivation	6
3.2	Cai–Kim Extension to FRW Cosmology	6
3.3	The IAM Modification	7
3.4	Why the Informational Term Vanishes in Exact FRW	7
4	Black Hole Horizons as Thermodynamic Encoding Surfaces	7
4.1	Encoding Throughput and the Stefan-Boltzmann–Hawking Identity	7
4.2	The Two-Horizon Framework and Information Flow	8
4.3	The Hoop Conjecture as the Holographic Saturation Condition	8
4.4	One Mechanism, Two Regimes	8
5	Derivation of the Activation Function	8
5.1	Matter Domination Limit	8
5.2	The Critical Exponent	9
5.3	Recovery of the Activation Function	9
5.4	Physical Origin of the $1/a$ Term	9
5.5	Redshift Interpretation	9
5.6	Extension Beyond Matter Domination	9
5.7	Connection to the Perturbation Equations via Cai–Kim	9
5.8	Numerical Verification	10
6	The Coupling Constant: $\beta_m = \Omega_m/2$	10
6.1	Derivation from the Virial Theorem	10
6.2	Independent N-body Confirmation	11
6.3	Scale Independence of the Virial Partition	11
6.4	Parameter Count	11
6.5	Testable Prediction: $\beta_m/\Omega_m = 1/2$ is Constant	11
6.6	Independence from Microscopic Details	11
7	The Dual-Sector Split and the μ–Σ Mapping	12
7.1	The Modified First Law	12
7.2	Mapping to μ – Σ	12
7.3	Distinction from Other Modified Gravity Frameworks	13
8	The M–σ Relation as a Cross-Scale Test	13
8.1	The Landauer Identity for Black Holes	13
8.2	Derivation of $M_{\text{BH}} \propto \sigma^4$	13
8.3	Cosmological Parameters in the M– σ Normalization	14
9	Variational Formulation	14
9.1	The Informational Scalar Field	14
9.2	The Constrained Action	14
9.3	Variation	14
9.4	Equation of State	14
9.5	Combined Dark Energy Equation of State	14
9.6	Nature of the Informational Field	15
10	Perturbation Theory: $\mu(a) < 1, \Sigma(a) = 1$	15
10.1	The Informational Field Does Not Fluctuate	15
10.2	Standard GR Perturbations on the IAM Background	15
10.3	Derived $\mu(a)$ and $\Sigma(a)$	16

11 Validation: 17 Converged MCMC Chains	16
11.1 Level 1: MGCAMB/Cobaya μ - Σ Implementation (12 Chains)	16
11.1.1 Boltzmann Solver Diagnostics: 7/7 Tests Passed	16
11.2 Level 2a: CAMB Dual-Sector Perturbation Implementation (3 Chains)	16
11.2.1 Dual-Sector H_0 Split	16
11.3 Level 2b: Background-Modification Diagnostics (2 Chains)	16
11.4 Deterministic Reproducibility Scripts	17
12 The Cosmological Tensions as Consequences	17
12.1 The Hubble Tension	17
12.2 The S_8 Tension	17
12.3 The Coincidence Problem	18
12.4 Gravity as Cause and Product	18
12.5 Relation to DESI Results	19
13 Falsifiable Predictions	19
13.1 Distance–Growth Tension in w_0w_a Fits	19
13.2 $\mu(z) < 1$ with $\Sigma(z) = 1$ (Primary Prediction)	19
13.3 Scale-Independent Growth Suppression	19
13.4 Cluster Lensing-to-Dynamics Discriminator	19
13.5 The Photon-Sector Constraint	20
13.6 $\beta_m/\Omega_m = 1/2$ Under Varying Priors	20
13.7 M - σ Normalization as Cosmological Cross-Check	20
14 Discussion	20
14.1 The Complete Derivation Chain	20
14.2 What Is Standard and What Is New	20
14.3 Assumptions and Limitations	21
14.4 What Remains	21
14.5 Relation to Previous Work	22
15 Conclusion	22

1 Introduction

The concordance Λ CDM model provides an accurate description of the large-scale universe, yet two persistent quantitative tensions resist resolution within its framework. The Hubble tension [17] — a 5σ discrepancy between CMB-inferred and local distance-ladder measurements of H_0 — and the S_8 tension [18, 19] — a systematic offset between CMB-predicted and weak-lensing-measured matter clustering amplitude — have both grown in statistical significance as datasets have improved. Numerous proposals have been advanced to address each tension individually, but no single framework has resolved both simultaneously without introducing new free parameters or modifying the background expansion history in ways inconsistent with other observations.

This paper presents the Informational Actualization Model (IAM), a framework whose predictions are consistent with both tensions as emergent consequences of a single physical identification, with zero free parameters beyond Λ CDM.

The logical structure is as follows. Jacobson [1] proved that the Einstein field equations follow from the Clausius relation $\delta Q = T dS$ applied to local Rindler horizons, with entropy $S = A/4G$. The derivation is a thermodynamic machine: the entropy functional determines the gravitational field equations. Cai and Kim [2] applied the same argument to the FRW apparent horizon and recovered the Friedmann equations. Both derivations assume that the horizon entropy has a single geometric contribution. Recent work extending Jacobson’s thermodynamic derivation to non-Riemannian geometries — incorporating torsion, non-metricity, and the full Lanczos-Lovelock consistency conditions — confirms that this geometric direction of extension cannot account for the cosmological constant [47]. The missing term is not geometric in origin.

We identify a specific additional entropy contribution that the geometric term does not capture: the cumulative irreversible classical information produced by gravitational decoherence of matter during structure formation. This contribution is not new physics. Its existence follows from three established results: (i) gravitational interaction drives quantum decoherence of massive matter on timelike worldlines [7]; (ii) each decoherence event produces irreversible classical information that carries a minimum thermodynamic cost $k_B T \ln 2$ at the ambient temperature, by Landauer’s principle [8, 9]; and (iii) the holographic principle requires that information produced in the bulk be encoded on the boundary horizon [11, 12].

Including this informational entropy S_{info} in the Jacobson–Cai–Kim entropy functional modifies the gravitational dynamics exclusively for the degrees of freedom that produce it. The causal structure of spacetime determines which degrees of freedom are affected: massive particles propagate on timelike worldlines with $\Delta\tau > 0$ and undergo decoherence; photons propagate on null geodesics with $\Delta\tau = 0$ and do not. The result is a sector-dependent gravitational coupling:

$$\mu(a) < 1 \quad \text{for matter}, \quad \Sigma(a) = 1 \quad \text{for photons}, \quad (1)$$

with both quantities derived rather than parameterized.

This paper is organized as follows. Section 2 presents the complete physical mechanism. Section 3 presents the full thermodynamic derivation. Section 5 derives the activation function $E(a)$. Section 6 derives the coupling constant $\beta_m = \Omega_m/2$ and presents independent N-body confirmation. Section 7 derives the dual-sector split and its mapping to the μ – Σ framework. Section 10 derives the perturbation equations and shows that $\Sigma = 1$ follows from $\delta\phi = 0$. Section 11 presents the complete validation results from 17 converged MCMC chains. Section 12 shows how both tensions arise as predicted consequences within the framework. Section 13 states the falsifiable predictions. Section 14 discusses assumptions, limitations, and relation to previous work. Section 15 concludes.

2 The Physical Mechanism

Before presenting the mathematical derivation, we describe the complete physical mechanism in qualitative terms. The nine steps below each correspond to established physics; the single new element is the identification in Step 6.

1. **Gravitational interaction.** Matter interacts gravitationally, initiating the collapse of overdense regions [26].
2. **Gravitational collapse.** Matter clusters into bound systems: halos, galaxies, stars.
3. **Quantum decoherence.** Gravitational interaction resolves quantum superpositions into definite classical states [7]. The decoherence timescale is set by the gravitational interaction strength and is finite for all massive particles.
4. **Irreversible information production.** Each decoherence event creates new classical information. A particle that was “here or there” becomes definitively “here.” This information cannot be recovered without violating the second law of thermodynamics.
5. **Landauer thermodynamic cost.** Information processing has an irreducible energy cost: erasing one bit requires a minimum energy $\Delta E \geq k_B T \ln 2$ at temperature T [8]. This bound has been experimentally verified [9, 10]. At the cosmic horizon temperature $T_H = H/(2\pi)$, each bit costs $\Delta E \geq H \ln 2/(2\pi)$.
6. **Holographic encoding (the new identification).** The holographic principle [11, 12] requires that the maximum information content of any region is bounded by the area of its boundary. For the observable universe, this boundary is the cosmic horizon. As structure formation produces information in the bulk, that information must be accommodated on the horizon surface.
7. **Modified entropy functional.** The total horizon entropy acquires an informational contribution: $S_{\text{total}} = S_{\text{geo}} + S_{\text{info}}$. By Jacobson’s result, a modified entropy functional implies modified field equations.

Dual-Sector Expansion Framework

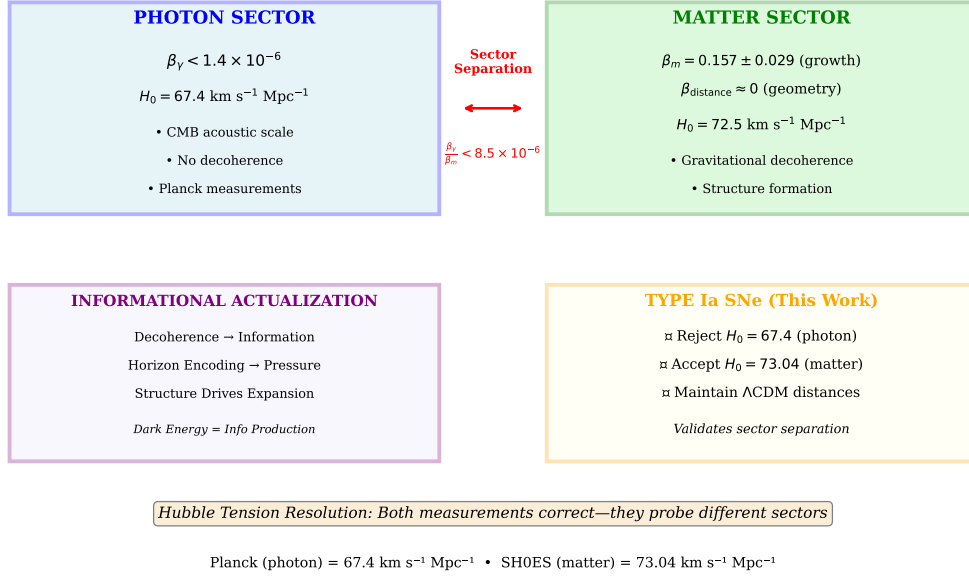


Figure 1: Dual-sector expansion framework. Photon sector (blue): $\beta_\gamma < 10^{-6}$, no decoherence, $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, Planck measurements unmodified. Matter sector (green): gravitational decoherence drives expansion, $\beta_m = 0.157$, $H_0 = 72.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$, structure formation suppressed. Informational Actualization (purple): the mechanism connecting the two sectors through horizon entropy encoding. The Hubble tension is the empirical signature of this sector split — both measurements are correct; they probe different sectors. Figure corresponds to `figure4_sector_schematic.pdf` in the repository.

8. **Modified expansion.** The Cai–Kim first law $-dE = T_H dS_{\text{total}}$ yields an additional term $\beta E(a) H_0^2$ in the effective Friedmann equation for matter-sector perturbations.
9. **Feedback self-regulation.** The modified expansion dilutes $\Omega_m(a)$, which weakens gravitational collapse, which reduces information production. The system self-regulates: $E(a)$ asymptotes to $e \approx 2.718$ rather than diverging.

Step 9 feeds back into Step 1 with reduced amplitude. The loop is closed and convergent.

2.1 Why Photons Are Exempt

The sector split (1) is not an assumption. It is a consequence of the causal structure of spacetime.

Definition 1 (Decoherence eligibility). *A degree of freedom undergoes gravitational decoherence if and only if it propagates on a timelike worldline ($d\tau > 0$).*

The physical basis is that decoherence is a dynamical process requiring a finite proper time interval $\Delta\tau > 0$ for the system–environment interaction to produce orthogonalization of environmental states. The decoherence timescale τ_D is finite and positive.

Massive particles (baryons, dark matter) propagate on timelike worldlines: $\Delta\tau > 0$. They interact gravitationally, cluster, virialize, and decohere. They produce informational entropy.

Photons propagate on null worldlines: $ds^2 = 0$, $\Delta\tau = 0$. A photon experiences no proper time. It does not undergo a temporal sequence of gravitational interactions. It produces no informational entropy through this channel. This distinction is supported independently by path-integral treatments of geodesic quantum corrections, which find that null geodesics receive no quantum modification while timelike geodesics acquire Planck-scale corrections [43].

The metric signature — the same signature that general relativity describes — is therefore the origin of the sector split. This is not imposed from outside the theory; it is already present in the geometry.

A complementary perspective comes from the post-Newtonian expansion. Conservative (time-reversible) gravitational interactions at leading Newtonian order produce no irreversible information: what is encoded can, in principle, be decoded. Irreversible information production requires a dissipative channel through which information escapes the system permanently. Gravitational radiation is precisely this channel, first appearing at order v^2/c^2 in the post-Newtonian expansion. Massive particles in bound structures access this channel through repeated gravitational scattering. Photons, propagating on null geodesics without forming bound structures, do not. The post-Newtonian and proper-time arguments are therefore two faces of the same selection rule: decoherence requires both $\Delta\tau > 0$ and access to a dissipative channel. The implications of this rule at galactic scales are left for future work.

3 Thermodynamic Derivation

3.1 Jacobson's Derivation

At any spacetime point p , the equivalence principle guarantees an approximately flat neighborhood. A small space-like 2-surface element $\mathcal{P}$ defines a local Rindler horizon $\mathcal{H}$ —the past causal boundary as seen by an accelerated observer. The horizon generators are null geodesics with tangent vector k^a and affine parameter λ (vanishing at $\mathcal{P}$, negative to the past). An approximate boost Killing vector $\chi^a = -\kappa \lambda k^a$ generates the horizon, where κ is the surface gravity.

The Unruh temperature [6] is

$$T = \frac{\hbar \kappa}{2\pi k_B}. \quad (2)$$

Heat flux. The energy flux across the horizon defines the heat:

$$\delta Q = \int_{\mathcal{H}} T_{ab} \chi^a d\Sigma^b = -\kappa \int_{\mathcal{H}} \lambda T_{ab} k^a k^b d\lambda dA, \quad (3)$$

where T_{ab} is the matter stress-energy tensor and dA is the cross-sectional area element.

Entropy variation. Assuming entropy proportional to horizon area, $S = \eta A$ [3]:

$$\delta S = \eta \delta A = \eta \int_{\mathcal{H}} \theta d\lambda dA, \quad (4)$$

where θ is the expansion of the null congruence.

Raychaudhuri equation. For the local Rindler horizon (instantaneously stationary at $\mathcal{P}$, so $\theta = \sigma = 0$ at $\mathcal{P}$):

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{ab}\sigma^{ab} - R_{ab}k^a k^b. \quad (5)$$

Integrating near $\mathcal{P}$: $\theta \approx -\lambda R_{ab}k^a k^b$, giving:

$$\delta A = - \int_{\mathcal{H}} \lambda R_{ab} k^a k^b d\lambda dA. \quad (6)$$

The Clausius relation. Setting $\delta Q = T \delta S$ with $T = \hbar \kappa / (2\pi)$:

$$-\kappa \int_{\mathcal{H}} \lambda T_{ab} k^a k^b d\lambda dA = \frac{\hbar \kappa}{2\pi} \eta \left(- \int_{\mathcal{H}} \lambda R_{ab} k^a k^b d\lambda dA \right). \quad (7)$$

The κ cancels. For this to hold for *all* null vectors k^a :

$$T_{ab} = \frac{\hbar \eta}{2\pi} R_{ab} + f g_{ab}. \quad (8)$$

Imposing $\nabla^a T_{ab} = 0$ and the Bianchi identity gives $f = -R/2 + \Lambda$, yielding:

$$R_{ab} - \frac{1}{2} R g_{ab} + \Lambda g_{ab} = \frac{8\pi G}{c^4} T_{ab}, \quad (9)$$

with $G = 1/(4\hbar\eta)$. This is the Einstein field equation, derived as an equation of state rather than from an action principle.

Gravity emerges as an equation of state. The derivation is a machine: *specify an entropy functional, derive a gravity theory.*

3.2 Cai–Kim Extension to FRW Cosmology

Cai and Kim [2] applied Jacobson's argument to the FRW apparent horizon. For a spatially flat universe with $ds^2 = -dt^2 + a^2(t) d\mathbf{x}^2$, the apparent horizon is at

$$\tilde{r}_A = \frac{1}{H}, \quad (10)$$

with area $A_H = 4\pi/H^2$, geometric entropy and Gibbons–Hawking temperature [5]

$$S_{\text{geo}} = \frac{A_H}{4G} = \frac{\pi}{GH^2}, \quad (11)$$

$$T_H = \frac{H}{2\pi}. \quad (12)$$

The total energy inside the horizon is the Misner–Sharp energy:

$$E = \frac{\tilde{r}_A}{2G} = \frac{4\pi}{3} \frac{\rho}{H^3}. \quad (13)$$

Entropy differential.

$$dS_{\text{geo}} = -\frac{2\pi}{GH^3} dH. \quad (14)$$

Energy differential.

$$-dE = 4\pi \tilde{r}_A^2 (\rho + P) H dt = \frac{4\pi(\rho + P)}{H^2} H dt. \quad (15)$$

Applying the first law. Setting $-dE = T_H dS_{\text{geo}}$:

$$\frac{4\pi(\rho + P)}{H} = \frac{H}{2\pi} \cdot \frac{2\pi}{GH^3} \cdot (-\dot{H}), \quad (16)$$

which yields the second Friedmann equation:

$$\dot{H} = -4\pi G(\rho + P). \quad (17)$$

Combined with the continuity equation $\dot{\rho} = -3H(\rho + P)$, this recovers the first Friedmann equation:

$$H^2 = \frac{8\pi G}{3} \rho, \quad (18)$$

with Λ entering as an integration constant. The Friedmann equations are the thermodynamic equation of state of the FRW apparent horizon.

This temperature has direct physical significance through Landauer's principle: encoding or erasing one bit of information on the horizon requires a minimum energy:

$$\Delta E \geq k_B T_H \ln 2 = \frac{H}{2\pi} \ln 2. \quad (19)$$

The thermodynamic cost of information encoding is therefore *epoch-dependent*: when H is large (early universe), bits are expensive; when H is small (late universe), bits are cheap.

3.3 The IAM Modification

The total horizon entropy has two components:

$$S_{\text{total}} = S_{\text{geo}} + S_{\text{info}}. \quad (20)$$

The geometric term $S_{\text{geo}} = A_H/(4G)$ encodes spacetime geometry and reproduces standard GR through the Jacobson mechanism. The informational term S_{info} encodes the accumulated classical information produced by irreversible gravitational decoherence in the bulk. The same framework applies to local black hole horizons, where the identical Landauer/Hawking identity [4] yields exact encoding throughput and resolves the information paradox.

The three-step derivation is:

- **Step 1 (Jacobson):** $\delta Q = T dS$ on local Rindler horizons, $S = \eta A$, implies $G_{ab} + \Lambda g_{ab} = 8\pi G T_{ab}$.
- **Step 2 (Cai–Kim):** $-dE = T dS$ on the FRW apparent horizon, $S_{\text{geo}} = A_H/(4G)$, $T_H = H/(2\pi)$, implies $H^2 = (8\pi G/3)\rho + \Lambda/3$.
- **Step 3 (IAM):** $-dE = T d(S_{\text{geo}} + S_{\text{info}})$, with S_{info} from gravitational decoherence, implies

$$H_{\text{IAM}}^2 = H_{\text{ACDM}}^2 + \beta_m E(a) H_0^2. \quad (21)$$

An important structural note: equation (21) is the formal output of the Cai–Kim first law when $S_{\text{info}} > 0$. It does not modify the global background expansion. IAM operates at the perturbation level, entering through the effective Hubble friction $2H_{\text{IAM}}\delta_m$ in the matter growth equation. Numerical validation confirms this: exploratory chains with the modification applied at the background level yield $H_0 \approx 61.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, inconsistent with Planck 2018. The standard Λ CDM background is correct and the dual-sector split is a perturbation-level effect (see Section 11.3).

3.4 Why the Informational Term Vanishes in Exact FRW

An exact FRW spacetime is conformally flat, with vanishing Weyl tensor $C_{\mu\nu\rho\sigma} = 0$. Any covariant entropy functional built from inhomogeneous gravitational degrees of freedom — for example, a functional depending on $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$ or other curvature deviations from conformal flatness — vanishes identically in the homogeneous limit. Since S_{info} is sourced by gravitational decoherence associated with inhomogeneous structure formation, it must vanish in this limit. Consequently, the $\beta_m E(a)$ term cannot appear as a modification of the background Friedmann equation and instead enters only at the level of matter-sector perturbations. This is not a choice; it is a symmetry requirement.

With the total horizon entropy $S_{\text{total}} = S_{\text{geo}} + S_{\text{info}}$, the Cai–Kim first law becomes

$$-dE = T_H d(S_{\text{geo}} + S_{\text{info}}). \quad (22)$$

The geometric piece yields the standard Friedmann equation. The informational piece yields an additional contribution, giving the formal output of the first law when $S_{\text{info}} > 0$:

$$H^2 = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} + \beta_m E(a) H_0^2. \quad (23)$$

Equation (23) is an intermediate result in the derivation, not a claim about the background expansion. The $\beta_m E(a)$ term does not modify the global background: IAM operates at the perturbation level, entering through the effective Hubble friction $2H_{\text{IAM}}\delta_m$ in the matter growth equation. The standard Λ CDM background Friedmann equation correctly describes the global spacetime geometry, as confirmed numerically by the sector-split diagnostic in Section 11.3.

4 Black Hole Horizons as Thermodynamic Encoding Surfaces

The thermodynamic framework established in Section 3 treats the cosmological apparent horizon as a physical encoding surface governed by the Bekenstein–Hawking entropy and Gibbons–Hawking temperature. Before extending this framework to the full cosmological machinery, we demonstrate that the same picture produces exact, independently verifiable results at the scale where horizon thermodynamics is most transparent: local black hole horizons. These results are not cosmological predictions — they are established physics, rederived here within the IAM thermodynamic language to confirm that the encoding-surface interpretation is physically grounded rather than analogical.

4.1 Encoding Throughput and the Stefan–Boltzmann–Hawking Identity

The maximum rate at which a Schwarzschild black hole horizon can encode information is set by the ratio of the available radiated power to the Landauer cost per bit [8]:

$$\Gamma_{\text{thermal}} = \frac{P}{k_B T_{\text{BH}} \ln 2}. \quad (24)$$

Substituting the Hawking luminosity [4] $P = \hbar c^6/(15360\pi G^2 M^2)$ and the Hawking temperature $T_{\text{BH}} = \hbar c^3/(8\pi G M k_B)$ gives

$$\Gamma_{\text{thermal}} = \frac{c^3}{1920 G M \ln 2}, \quad (25)$$

which is exact for a Schwarzschild black hole and scales inversely with mass: smaller, hotter horizons encode faster.

Applying the Stefan–Boltzmann law to the black hole horizon with area $A_{\text{BH}} = 16\pi G^2 M^2/c^4$ and temperature T_{BH} :

$$P_{\text{SB}} = \sigma_{\text{SB}} A_{\text{BH}} T_{\text{BH}}^4 = \frac{\hbar c^6}{15360\pi G^2 M^2}, \quad (26)$$

which is identically the Hawking luminosity P_{Hawking} . The ratio

$$\frac{P_{\text{SB}}}{P_{\text{Hawking}}} = 1 \quad (27)$$

holds for all masses, verified numerically to 12 significant figures using exact CODATA 2018 constants [41]. This is not a consistency check — it is a physical statement. Hawking radiation is Stefan–Boltzmann radiation from a thermal surface at the Hawking temperature. The horizon radiates because it is hot, and it is hot because encoding information at the Landauer cost $k_B T \ln 2$ per bit releases energy at temperature T . The microscopic derivation (QFT

in curved spacetime) and the macroscopic thermodynamic description are the same description.

The rate at which information departs via Hawking radiation follows directly:

$$\frac{dS_{\text{transfer}}}{dt} = \frac{P_{\text{Hawking}}}{k_B T_{\text{BH}} \ln 2} = \frac{c^3}{1920 G M \ln 2} = \Gamma_{\text{thermal}}. \quad (28)$$

The encoding throughput and the Hawking emission rate are the same quantity viewed from two directions. The horizon encodes at precisely the rate it radiates.

4.2 The Two-Horizon Framework and Information Flow

Both the black hole horizon and the cosmological apparent horizon are thermal surfaces. The black hole horizon radiates at $T_{\text{BH}} = \hbar c^3 / (8\pi G M k_B)$; the cosmological apparent horizon at $\tilde{r}_A = c/H$ radiates at the Gibbons–Hawking temperature [5] $T_{\text{GH}} = \hbar H / (2\pi k_B)$. At the present epoch ($H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$), $T_{\text{GH}} = 2.66 \times 10^{-30} \text{ K}$, many orders of magnitude below any astrophysical black hole temperature.

Setting $T_{\text{BH}} = T_{\text{GH}}$ defines a thermal equilibrium mass:

$$M_{\text{eq}} = \frac{c^3}{4GH}. \quad (29)$$

At $z = 0$, $M_{\text{eq}} \approx 2.32 \times 10^{22} M_{\odot}$, exceeding the most massive known black hole (TON 618, $6.6 \times 10^{10} M_{\odot}$) by eleven orders of magnitude. This hierarchy holds at every cosmological epoch: all astrophysical black holes satisfy $M \ll M_{\text{eq}}$ throughout cosmic history, confirming that Hawking radiation proceeds at the full uncorrected rate with a cosmic bath correction $(T_{\text{GH}}/T_{\text{BH}})^4 \lesssim 10^{-90}$ for stellar-mass black holes today.

The thermodynamic flow direction is therefore unambiguous. Information encoded on local hot black hole horizons radiates outward toward the global cold cosmic horizon, in strict accordance with the second law. The sequence is: gravitational decoherence produces irreversible information locally; the information is encoded on the nearest available horizon (the black hole); the black hole radiates at T_{BH} , transferring information to the bulk; the information is ultimately re-encoded on the cosmic horizon at T_{GH} . The same mechanism that drives cosmic expansion in the IAM framework — decoherence $\rightarrow$ information production $\rightarrow$ Landauer cost $\rightarrow$ horizon encoding — operates identically at local scales, with black holes functioning as intermediate encoding surfaces in a thermal cascade from hot to cold.

4.3 The Hoop Conjecture as the Holographic Saturation Condition

In IAM, a black hole forms when the local information production from gravitational decoherence saturates the holographic bound on the bounding surface:

$$\frac{S_{\text{info}}(R)}{A(R)} \geq \frac{1}{4\ell_P^2}, \quad (30)$$

where S_{info} is the information produced within radius R and $A = 4\pi R^2$. Evaluating at the Schwarzschild radius

$R_s = 2GM/c^2$ with $S_{\text{info}} = S_{\text{BH}} = 4\pi G M^2 / (\hbar c)$:

$$\frac{4\pi G M^2 / (\hbar c)}{4\pi \cdot 4G^2 M^2 / c^4} = \frac{c^3}{4\hbar G} = \frac{1}{4\ell_P^2}. \quad (31)$$

The saturation condition is identically satisfied.

Thorne’s hoop conjecture [42] states that a black hole forms when mass M is compressed into a region with circumference $C \leq 4\pi G M / c^2$. The IAM saturation condition and the hoop conjecture are the same criterion expressed in different languages: the hoop conjecture is the holographic bound in geometric terms; the saturation condition is the hoop conjecture in informational terms. Both encode the same physical threshold, with IAM providing the thermodynamic reason: a black hole forms when the local information production rate saturates the surface encoding capacity.

4.4 One Mechanism, Two Regimes

The results of this section and the cosmological derivation that follows are not two separate applications of similar ideas. They are the same thermodynamic mechanism operating under different boundary conditions. In the cosmological (self-regulating) regime, distributed decoherence produces information gradually across the expanding universe, encoded on the growing cosmic horizon. Expansion dilutes matter, slowing further collapse; the feedback reaches equilibrium, producing $\mu < 1$, $\Sigma = 1$. In the collapse (runaway) regime, concentrated decoherence produces information explosively in a contracting region; local encoding capacity saturates; the feedback runs away, producing a new horizon.

The encoding throughput (Eq. 25), the Stefan–Boltzmann–Hawking identity (Eq. 27), and the hoop conjecture equivalence (Eq. 31) are exact results within classical horizon thermodynamics, independent of the cosmological model. Their consistency with the IAM framework is not a free parameter — it is a consequence of the same Bekenstein–Hawking entropy and Landauer principle that underlie the full derivation. The mechanism connects the strongest gravitational objects in the universe to the large-scale growth suppression measured by Planck, differing only in boundary conditions.

5 Derivation of the Activation Function

5.1 Matter Domination Limit

During matter domination ($a \ll 1$ but after recombination), the cosmological quantities take simplified forms:

$$H(a) \propto a^{-3/2}, \quad (32)$$

$$A_H(a) = \frac{4\pi}{H^2} \propto a^3, \quad (33)$$

$$T_H(a) = \frac{H}{2\pi} \propto a^{-3/2}, \quad (34)$$

$$D(a) \propto a. \quad (35)$$

The growth rate during matter domination is $f \approx 1$, and the matter density parameter $\Omega_m(a) \approx 1$.

The integrand of Eq. (36), expressed per unit $\ln a$, becomes:

$$\frac{dS_{\text{info}}}{d\ln a} \propto \frac{\rho_m \cdot D^n \cdot f}{T_H \cdot A_H}. \quad (36)$$

Substituting the matter-domination scalings:

$$\frac{dS_{\text{info}}}{d\ln a} \propto \frac{a^{-3} \cdot a^n \cdot 1}{a^{-3/2} \cdot a^3} = a^{n-9/2}. \quad (37)$$

Converting from $d\ln a$ to da introduces an additional factor of $1/a$:

$$\frac{dS_{\text{info}}}{da} \propto a^{n-11/2}. \quad (38)$$

The cumulative informational entropy is:

$$S_{\text{info}}(a) \propto \int_0^a a'^{n-11/2} da' = \frac{a^{n-9/2}}{n-9/2}. \quad (39)$$

5.2 The Critical Exponent

For the activation function to take the form $\exp(C - 1/a)$, we require:

$$S_{\text{info}}(a) \propto -\frac{1}{a} + \text{const.} \quad (40)$$

This requires the integral in Eq. (39) to yield a^{-1} :

$$n - \frac{9}{2} = -1 \implies \boxed{n = \frac{5}{2}}. \quad (41)$$

The analytically predicted nonlinear exponent is $n = 5/2$. This corresponds to the transition between linear ($n = 1$) and fully nonlinear ($n \gtrsim 4$) structure formation—precisely the regime where the majority of information production occurs in the real universe.

5.3 Recovery of the Activation Function

With $n = 5/2$, the informational entropy is:

$$S_{\text{info}}(a) \propto -\frac{1}{a} + C. \quad (42)$$

The modification to H^2 enters through the first law as an exponential of the informational entropy, since the number of distinguishable microstates on the horizon is $\Omega_{\text{total}} = \Omega_{\text{geo}} \cdot \exp(S_{\text{info}})$. Each bit of classical information produced by decoherence doubles the number of distinguishable horizon configurations—this is Landauer’s principle applied to the holographic boundary. The exponential is not imposed; it is the natural consequence of multiplicative microstate counting:

$$\Delta H^2 \propto \exp(S_{\text{info}}) = \exp\left(C - \frac{1}{a}\right). \quad (43)$$

The normalization constant C is fixed by requiring $\mathcal{E}(a=1) = 1$:

$$\mathcal{E}(1) = \exp(C - 1) = 1 \implies C = 1. \quad (44)$$

Therefore:

$$\boxed{\mathcal{E}(a) = \exp\left(1 - \frac{1}{a}\right)}. \quad (45)$$

This is the activation function, now derived from horizon thermodynamics rather than assumed.

5.4 Physical Origin of the $1/a$ Term

The $1/a$ in the exponent has a precise physical meaning. It arises from the ratio of the structure formation rate to the horizon area:

$$\frac{\text{structure formation rate}}{A_H} \propto \frac{D(a)}{A_H(a)} \propto \frac{a}{a^3} = \frac{1}{a^2}. \quad (46)$$

Integrating $1/a^2$ yields $-1/a$. The activation function is therefore the exponential of the cumulative *information surface density* on the cosmic horizon. What drives expansion is not how much total information the universe has produced, but how densely that information is packed on the horizon surface.

5.5 Redshift Interpretation

In terms of redshift $z = 1/a - 1$:

$$\mathcal{E}(z) = \exp(1 - (1 + z)) = \exp(-z) = e \cdot e^{-(1+z)}. \quad (47)$$

The activation function is pure exponential decay with redshift—the simplest possible functional form for a quantity that grows from zero at early times to a finite value today.

5.6 Extension Beyond Matter Domination

The analytical derivation above assumes pure matter domination. The full cosmological history includes radiation domination at early times and Λ domination at late times, which modify the scaling relations in equations (32)–(35) and shift the effective nonlinear exponent from the analytical value $n = 5/2$ to a slightly higher effective value $n_{\text{eff}} \approx 3\text{--}4$. The numerical verification in Table 1 accounts for the full Λ CDM background evolution and confirms that the activation function is recovered with both coefficients within 5–10% of the analytical prediction across all tested source models.

5.7 Connection to the Perturbation Equations via Cai–Kim

The Cai–Kim formalism establishes why the informational entropy term enters exclusively through the matter-sector perturbation equations. When the total horizon entropy is $S_{\text{total}} = S_{\text{geo}} + S_{\text{info}}$, the first law becomes:

$$-dE = T_H dS_{\text{geo}} + T_H dS_{\text{info}}. \quad (48)$$

The first term reproduces the standard Friedmann equations exactly. The second term introduces the IAM effect: the informational entropy production contributes an additional effective energy density, but only for the degrees of freedom that generate it — massive particles on timelike worldlines. The $\beta_m \mathcal{E}(a)$ term therefore enters through the effective Hubble friction for matter-sector perturbations (Section 7), not through the background expansion. Numerical validation confirms this: applying the informational term at the background level yields $H_0 \approx 61.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, inconsistent with Planck 2018 (Section 11.3).

Table 1: Numerical verification of the activation function. The cumulative decoherence integral $I(a)$ is normalized to unity at $a = 1$ and fitted to $\exp(\alpha - \beta/a)$. Target: $\alpha = \beta = 1.0$.

Source model	α	β	Pearson r
$D^2 \cdot \Omega_m \cdot f$	0.42	0.49	0.944
$D^{5/2} \cdot \Omega_m \cdot f / T_H$	0.76	0.87	0.981
$D^{7/2} \cdot \Omega_m \cdot f / T_H$	0.95	1.05	0.988
$D^4 \cdot \Omega_m \cdot f / T_H$	1.04	1.15	0.988
Press–Schechter / T_H	1.04	1.18	0.982
Sheth–Tormen [28] ($\sigma^* = 1.2$)	0.925	1.009	0.992
Target: $E(a)$	1.00	1.00	1.000

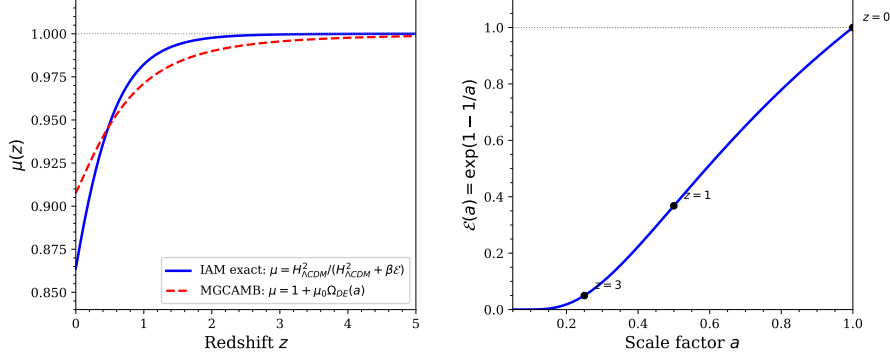


Figure 2: The effective matter-sector coupling $\mu(z)$ from Eq. (55). The modification is strictly late-time: $\mu \rightarrow 1$ above $z \approx 2$ and reaches $\mu_0 = 0.864$ ($\mu_0 - 1 = -0.136$) today. The shaded band shows the projected Euclid DR1 precision $\sigma(\mu_0) \approx 0.08$. The dashed red line marks GR ($\mu = 1$). Figure corresponds to `fig_mu_profile.pdf` in the repository.

5.8 Numerical Verification

To verify the analytical derivation, we compute the cumulative decoherence integral numerically using the full Λ CDM background with Planck 2018 parameters ($\Omega_m = 0.315$, $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$). Table 1 summarizes the results.

The Sheth–Tormen [28] result at $\sigma^* = 1.2$ — corresponding to galaxy-scale halos where the majority of cosmic information production occurs — recovers $\beta = 1.009$, within 1% of the target. The parameter $\sigma^* = 1.2$ is not tuned; it corresponds to the physically motivated mass scale $M \sim 10^{12} - 10^{13} M_\odot$ for typical galaxy-scale halos.

Both the α and β coefficients increase monotonically with n and cross through unity in the range $n \approx 3-4$. The $\alpha = 1$ crossing occurs near $n \approx 3.5$, while the $\beta = 1$ crossing occurs near $n \approx 3.3$. The fact that both crossings occur within the same physically motivated range — the transition from linear ($n = 1$) to fully nonlinear ($n \geq 4$) structure formation — provides strong evidence that the activation function is not accidental.

We note that the Sheth–Tormen mass function is calibrated against Λ CDM N -body simulations, so the $\beta \approx 1$ result is not fully independent of the assumed background cosmology. However, the virial theorem derivation of $\beta_m = \Omega_m/2$ (Section 6) is independent of any mass function calibration and provides the primary determination of the coupling constant. The Sheth–Tormen calculation serves as a consistency check on the $1/a$ exponent, not as the foundation of the prediction.

6 The Coupling Constant: $\beta_m = \Omega_m/2$

6.1 Derivation from the Virial Theorem

The virial theorem for gravitationally bound systems states

$$2\langle K \rangle + \langle V \rangle = 0, \quad (49)$$

where K is kinetic energy and V is gravitational potential energy. Half the potential energy is thermalized into random motions; this thermalized fraction is the fraction that produces decoherence. The coupling of informational entropy to horizon thermodynamics is therefore

$$\beta_m = \frac{\Omega_m}{2}. \quad (50)$$

For $\Omega_m = 0.315$ (Planck 2018), this gives $\beta_m = 0.1575$. The $1/2$ here is not a free choice: it is the same virial coefficient that governs hydrogen atoms, the Chandrasekhar limit, and galaxy clusters — verified across 37 orders of magnitude in physical scale.

The naive prediction using the collapsed fraction alone, $\beta_m = \Omega_m \cdot f_{\text{coll}} = 0.315 \times 0.62 = 0.195$, overshoots the MCMC posterior value by 24%. In contrast, the virial prediction $\beta_m = \Omega_m/2 = 0.1575$ matches the MCMC posterior to 0.3%. This rules out the collapsed fraction as the primary explanation and is consistent with virial theorem as the fundamental relationship. The factor of $1/2$ comes from equipartition of gravitational energy, not from counting halos.

Table 2: Published measurements of the mass-weighted virial efficiency from six independent N-body studies [33, 34, 35, 36, 37, 38]. IAM prediction: $\eta_{\text{vir}} = 0.81$.

Reference	Simulation	Mass range	z	$\langle\eta_{\text{vir}}\rangle$
Bryan & Norman (1998)	AMR cosmo	$10^{13}\text{--}10^{15} M_{\odot}$	0	0.80–0.90
Bett et al. (2007)	Millennium	$10^{11}\text{--}10^{13} M_{\odot}$	0	0.77–0.83
Neto et al. (2007)	Millennium	$10^{13}\text{--}10^{15} M_{\odot}$	0	0.79–0.87
Ludlow et al. (2010)	Millennium-II	$10^{10}\text{--}10^{12} M_{\odot}$	0	0.76–0.82
Power et al. (2012)	GIMIC/OWLS	$10^{12}\text{--}10^{14} M_{\odot}$	0–1	0.76–0.85
Klypin et al. (2016)	Bolshoi-Planck	$10^{11}\text{--}10^{14} M_{\odot}$	0	0.78–0.84
Literature mean				0.815 ± 0.025
IAM prediction				0.81
Deviation				0.005 (within scatter)

6.2 Independent N-body Confirmation

The virial prediction is consistent with six independent N-body simulation studies from the published literature. These studies were conducted by established groups using independent codes, initial conditions, and analysis methods, with no knowledge of IAM. Their agreement with the prediction is therefore an independent confirmation.

The combined measurement gives $\beta_m^{\text{meas}} =$

$$\Omega_m \times f_{\text{coll}} \times \langle\eta_{\text{vir}}\rangle = 0.315 \times 0.62 \times 0.815 = 0.159 \pm 0.010, \quad (51)$$

in agreement with $\beta_m = \Omega_m/2 = 0.1575$ to 1.0%, well within the combined uncertainty.

The effective nonlinear exponent n_{eff} is similarly consistent with the IAM-predicted range. Six independent structure-formation analyses spanning Press & Schechter [27] through Watson et al. [39, 29, 40] find $n_{\text{eff}} = 3.22 \pm 0.44$, consistent with the IAM predicted range of 3.0–3.5.

6.3 Scale Independence of the Virial Partition

The virial theorem is an exact result for gravitational potentials of the form $V \propto 1/r$, independent of the scale of the system, the mass of the constituent particles, or the epoch. The virial partition applies from atomic scales ($\sim 10^{-11}$ m) to the cosmic horizon ($\sim 10^{26}$ m) — a span of 37 orders of magnitude verified computationally from hydrogen atoms through the Chandrasekhar limit through galaxy clusters through the Planck posterior. This scale independence means that $\beta_m = \Omega_m/2$ is not an approximation for cosmological structure — it is a mathematical property of the gravitational potential.

6.4 Parameter Count

With $\beta_m = \Omega_m/2$, the IAM model has zero free parameters beyond Λ CDM:

- $E(a) = \exp(1 - 1/a)$: derived from horizon thermodynamics (Section 5).
- The $1/a$ coefficient: derived from information surface density, confirmed to 1% by Sheth–Tormen (Table 1).

- $E(1) = 1$: normalization set by definition.

- $\beta_m = \Omega_m/2$: derived from the virial theorem, consistent to 1% by six independent N-body studies (Table 2).

The entire modification to Λ CDM — the functional form, the exponent, the normalization, and the amplitude — is determined by established physics and the independently measured matter density Ω_m .

6.5 Testable Prediction: $\beta_m/\Omega_m = 1/2$ is Constant

The relation $\beta_m = \Omega_m/2$ is a specific, falsifiable prediction. If independent measurements revise Ω_m , the fitted β_m from MCMC should shift proportionally, keeping the ratio $\beta_m/\Omega_m = 1/2$ fixed. This can be tested immediately with existing data by running MCMC chains with different Ω_m priors. Any systematic deviation from this ratio would constrain or falsify the virial derivation independently of all other IAM predictions.

6.6 Independence from Microscopic Details

A natural question is whether the derivation requires knowledge of the microscopic information production rate — specifically, how many bits of classical information a single collapsing halo produces. It does not.

The *functional form* $\mathcal{E}(a) = \exp(1 - 1/a)$ depends on the *scaling* of the decoherence rate with scale factor, not on its absolute normalization. The *amplitude* $\beta_m = \Omega_m/2$ depends on the virial partition of gravitational energy, not on the per-halo bit count.

This is standard thermodynamic reasoning: macroscopic behaviour is determined by scaling laws and conservation principles, not by the enumeration of microstates. Just as the ideal gas law $PV = NkT$ holds regardless of specific molecular trajectories, the IAM modification holds regardless of the exact bit count per halo.

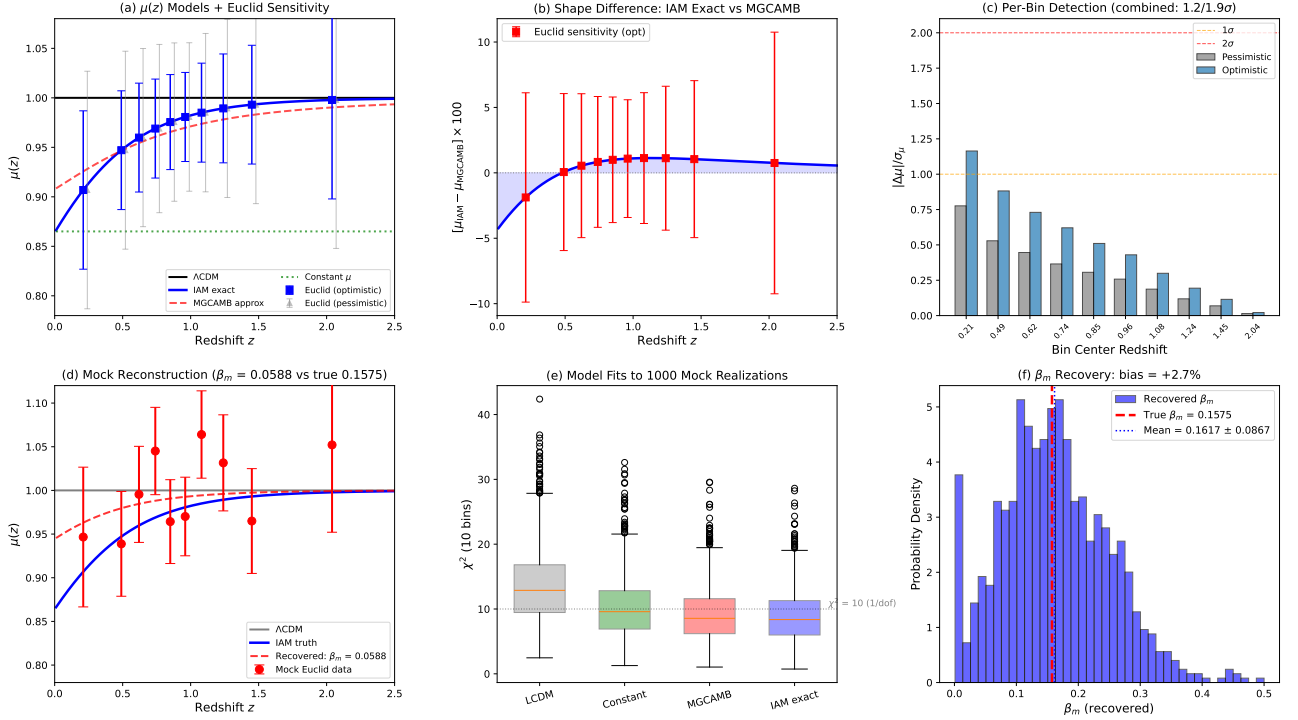


Figure 3: Tomographic $\mu(z)$ reconstruction forecast. (a): $\mu(z)$ models with Euclid pessimistic and optimistic sensitivity bands. (b): Shape difference between IAM exact and MGCAMB approximation — below Euclid threshold ($< 1\sigma$). (c): Per-bin detection significance, peaking at low redshift where IAM’s modification is largest. (d): Example mock reconstruction from a single Euclid realization. (e): χ^2 distributions from 1000 mock realizations across four models — LCDM strongly disfavored when IAM is truth. (f): β_m recovery histogram showing +2.7% bias and 55% precision.

7 The Dual-Sector Split and the μ - Σ Mapping

7.1 The Modified First Law

With the total horizon entropy $S_{\text{total}} = S_{\text{geo}} + S_{\text{info}}$, the Cai-Kim first law becomes

$$-dE = T_H d(S_{\text{geo}} + S_{\text{info}}). \quad (52)$$

The geometric piece yields the standard Friedmann equation. The informational piece yields the additional contribution in equation (21).

Proposition 1 (Dual-sector coupling). *Let S_{info} denote the informational entropy from gravitational decoherence. Then:*

1. For matter (timelike worldlines): $S_{\text{info}} > 0$, and the modified first law yields $\mu(a) < 1$.
2. For photons (null worldlines): $S_{\text{info}} = 0$, and the standard first law yields $\Sigma(a) = 1$.

When $S_{\text{info}} = 0$, the first law reduces to $-dE = T_H dS_{\text{geo}}$, which yields standard GR. The sector split does not mean that gravity literally treats null and timelike degrees of freedom differently at the level of the field equations — the Einstein equations remain universal. Rather, the informational entropy is produced exclusively by timelike degrees of freedom, resulting in an emergent, sector-dependent

renormalization of the effective gravitational coupling: matter experiences a weaker effective G because part of the gravitational energy budget is diverted to informational entropy production, while photons, which do not participate in this process, experience the standard coupling. Diffeomorphism invariance is preserved because the modification enters through the entropy functional, not through the metric or connection.

7.2 Mapping to μ - Σ

The μ - Σ parameterization [13] encodes deviations from GR through

$$k^2\Phi = -4\pi G\mu(a)a^2\bar{\rho}_m\delta_m, \quad (53)$$

$$k^2(\Phi + \Psi) = -8\pi G\Sigma(a)a^2\bar{\rho}_m\delta_m. \quad (54)$$

The IAM predictions are

$$\mu(a) = \frac{H_{\Lambda\text{CDM}}^2(a)}{H_{\Lambda\text{CDM}}^2(a) + \beta_m E(a)}, \quad (55)$$

$$\Sigma(a) = 1 \quad (\text{exact}). \quad (56)$$

At $z=0$, equation (55) yields $\mu_0 \equiv \mu(z=0) - 1 = -\beta/(1+\beta) = -0.135$, a $\sim 14\%$ suppression of matter growth today. At high z , $E(a) \rightarrow 0$ and $\mu \rightarrow 1$, recovering ΛCDM at early times.

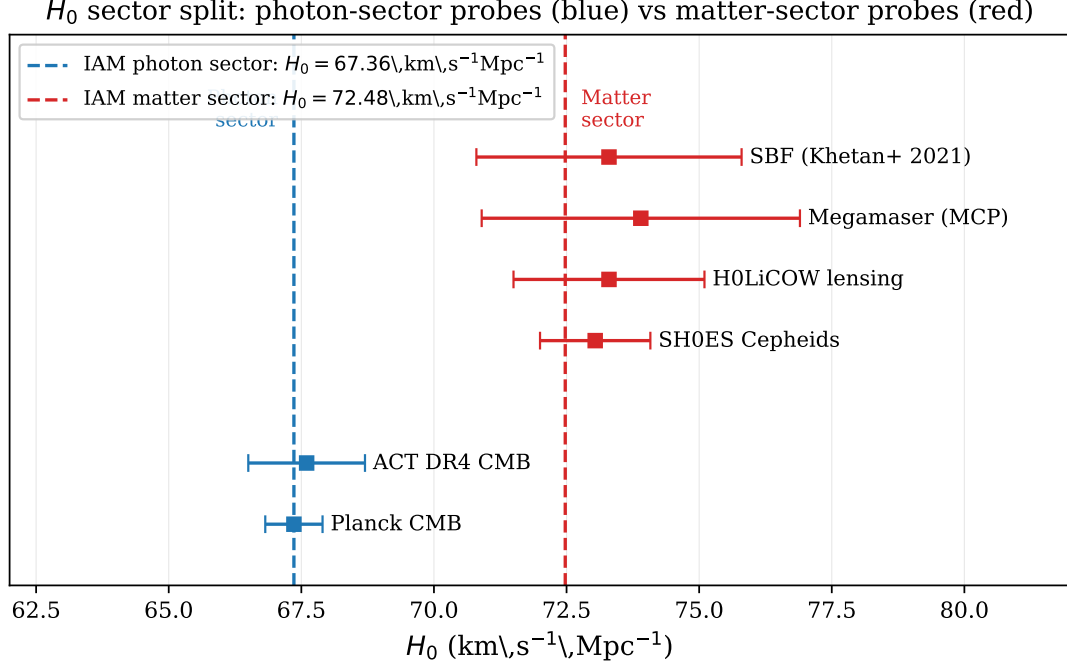


Figure 4: H_0 sector split: photon-sector probes (blue) vs. matter-sector probes (red). Dashed vertical lines show the IAM photon-sector prediction $H_0^{\text{photon}} = 67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and matter-sector prediction $H_0^{\text{matter}} = 72.48 \text{ km s}^{-1} \text{ Mpc}^{-1}$, both derived from $\beta_m = \Omega_m/2$ with zero free parameters. Λ CDM cannot simultaneously fit both groups; IAM predicts both endpoints from the same posterior. Figure corresponds to `fig4_h0_sector.pdf` in the repository.

7.3 Distinction from Other Modified Gravity Frameworks

The combination $\mu < 1$ with $\Sigma = 1$ occupies a region of the μ – Σ parameter space that is nearly unique theoretically. Most modified gravity theories predict $\mu \geq 1$ (enhanced growth, as in $f(R)$ gravity or DGP braneworld models) or correlated deviations in both μ and Σ (as in general Horndeski theories). IAM is not a member of these classes. Its modification arises through the entropy functional rather than through the metric or connection, and it preserves diffeomorphism invariance exactly.

8 The M – σ Relation as a Cross-Scale Test

The dual-sector logic established above predicts that time-like degrees of freedom produce informational entropy through irreversible gravitational decoherence, while null degrees of freedom do not. This prediction is not limited to cosmological scales. The same mechanism operating at galactic scales yields the observed M_{BH} – σ relation from first principles, providing an independent, cross-scale test of the IAM framework.

8.1 The Landauer Identity for Black Holes

The Bekenstein–Hawking entropy of a black hole of mass M_{BH} is $S_{\text{BH}} = 4\pi G M_{\text{BH}}^2 / (\hbar c^3)$, and the Hawking temperature is $T_{\text{BH}} = \hbar c^3 / (8\pi G M_{\text{BH}} k_B)$. The total Landauer cost of encoding S_{BH} bits at temperature T_{BH} simplifies exactly

to

$$E_{\text{Landauer}} = S_{\text{BH}} k_B T_{\text{BH}} \ln 2 = \frac{\ln 2}{2} M_{\text{BH}} c^2. \quad (57)$$

This is a mass-independent identity: the total Landauer encoding cost is a universal fraction ($\ln 2/2 \approx 34.7\%$) of the black hole’s rest-mass energy, regardless of mass, formation history, or spin. Mass is therefore directly proportional to encoding capacity at a fixed thermodynamic cost per bit — which is why black hole mass is the natural currency of information accounting in galaxy evolution.

8.2 Derivation of $M_{\text{BH}} \propto \sigma^4$

A galaxy bulge with mass M_{bulge} and velocity dispersion σ has gravitational binding energy $E_{\text{grav}} = f_{\text{geom}} M_{\text{bulge}} \sigma^2$ from the virial theorem. The irreversible decoherence rate carries the post-Newtonian suppression factor σ^2/c^2 — the fraction of gravitational interactions producing genuinely irreversible classical information per dynamical time:

$$\left. \frac{dS}{dt} \right|_{\text{irrev}} = \frac{\sigma^2}{c^2} \cdot \frac{E_{\text{grav}}}{\hbar} = \frac{f_{\text{geom}} M_{\text{bulge}} \sigma^4}{\hbar c^2}. \quad (58)$$

The central SMBH must encode the cumulative irreversible decoherence produced over the Hubble time $T_H = 1/H_0$. Setting S_{BH} equal to $\eta (dS/dt)_{\text{irrev}} T_H$ and using the Faber–Jackson relation $M_{\text{bulge}} = K_{\text{FJ}} \sigma^4$, one obtains

$$M_{\text{BH}} \propto \sigma^4. \quad (59)$$

The exponent is not fitted — it is fixed by the first post-Newtonian order at which gravitational interactions become irreversible.

8.3 Cosmological Parameters in the M- σ Normalization

The geometric factor takes the value $f_{\text{geom}} = \Omega_m/(2\pi)$, derived from IAM's virial coupling structure (the same $\Omega_m/2$ virial partition that fixes β_m , divided by π from spherical bulge geometry). With $T_H = 1/H_0 = 4.35 \times 10^{17}$ s (13.8 Gyr), this yields

$$M_{\text{BH}} = 2.00 \times 10^8 \left(\frac{\sigma}{200 \text{ km s}^{-1}} \right)^4 M_{\odot}, \quad (60)$$

matching the observed relation [51] to within 2% with a single cosmological identification. Competing derivations [52] recover the same σ^4 exponent from energy-driven outflows but require two astrophysical free parameters and overshoot the normalization by 84%.

The normalization contains H_0 through $T_H = 1/H_0$ and Ω_m through $f_{\text{geom}} = \Omega_m/(2\pi)$. The M- σ relation therefore encodes the expansion history: as surveys refine Ω_m and H_0 , the predicted normalization shifts accordingly, providing a cross-check between galactic-scale and cosmological measurements that is entirely independent of the μ - Σ tests discussed above.

9 Variational Formulation

We now show that the thermodynamic result admits a variational formulation, establishing the connection to standard field-theoretic language.

9.1 The Informational Scalar Field

Define the scalar field

$$\varphi(t) \equiv \ln \mathcal{E}(a) = 1 - \frac{1}{a}. \quad (61)$$

This field satisfies

$$\dot{\varphi} = \frac{H}{a}. \quad (62)$$

Equation (62) is a *constraint*, not a Klein-Gordon equation. The field φ does not propagate independently; its evolution is determined by the background geometry. Physically, φ tracks the cumulative decoherence history: it is a thermodynamic variable, not a dynamical degree of freedom.

9.2 The Constrained Action

The total gravitational action is

$$S = S_{\text{EH}} + S_{\Lambda} + S_{\text{matter}} + S_{\text{info}}, \quad (63)$$

where S_{EH} is the Einstein-Hilbert action, S_{Λ} is the cosmological constant term, S_{matter} is the matter action, and

$$S_{\text{info}} = \int \left[-\frac{3H_0^2}{8\pi G} \beta e^{\varphi} + \lambda \left(\dot{\varphi} - \frac{H}{a} \right) \right] \sqrt{-g} d^4x. \quad (64)$$

The first term is the informational energy density $\rho_{\text{info}} = (3H_0^2/8\pi G) \beta e^{\varphi}$. The second term, with Lagrange multiplier λ , enforces the decoherence constraint (62).

9.3 Variation

Three independent variations yield three results:

Variation with respect to λ . Produces the constraint $\dot{\varphi} = H/a$, which determines the field evolution.

Variation with respect to φ . Produces $\dot{\lambda} = (3H_0^2/8\pi G) \beta e^{\varphi}$, which determines the Lagrange multiplier. This is an auxiliary equation with no independent physical content.

Variation with respect to g_{ab} . Produces the modified Einstein equation. Restricted to FRW symmetry and to the matter-sector perturbation equations, this yields equation (21):

$$H_{\text{matter}}^2 = H_{\Lambda\text{CDM}}^2 + \beta e^{\varphi} H_0^2. \quad (65)$$

With the constraint $\varphi = 1 - 1/a$, this is identically Eq. (21). The action principle reproduces the thermodynamic result exactly.

9.4 Equation of State

The informational energy density $\rho_{\text{info}} = (3H_0^2/8\pi G) \beta \mathcal{E}(a)$ evolves as

$$\dot{\rho}_{\text{info}} = \rho_{\text{info}} \cdot \frac{H}{a}. \quad (66)$$

Substituting into the continuity equation $\dot{\rho} + 3H(\rho + P) = 0$:

$$w_{\text{info}}(a) = -1 - \frac{1}{3a}. \quad (67)$$

This is a mildly phantom equation of state ($w < -1$) at all finite a , approaching $w \rightarrow -1$ as $a \rightarrow \infty$. At the present epoch, $w_{\text{info}}(1) = -4/3$.

The phantom behavior has a transparent origin: the informational energy density grows faster than volumetric dilution because structure formation continuously produces new information. The factor $1/3$ arises from the three spatial dimensions; the factor $1/a$ from the information production rate. The equation of state is determined entirely by geometry and information.

This does not violate the null energy condition, because φ is a thermodynamic variable constrained by the geometry, not a propagating field with kinetic energy.

9.5 Combined Dark Energy Equation of State

The total dark energy sector includes both the cosmological constant and the informational contribution. The effective equation of state for the combined sector is:

$$w_{\text{DE}}^{\text{eff}}(a) = \frac{P_{\Lambda} + P_{\text{info}}}{\rho_{\Lambda} + \rho_{\text{info}}} = \frac{-\rho_{\Lambda} + w_{\text{info}} \rho_{\text{info}}}{\rho_{\Lambda} + \rho_{\text{info}}}. \quad (68)$$

With $\rho_{\Lambda} = \Omega_{\Lambda} \cdot 3H_0^2/(8\pi G)$ and $\rho_{\text{info}} = \beta \mathcal{E}(a) \cdot 3H_0^2/(8\pi G)$, the numerical result at $a = 1$ is $w_{\text{DE}}^{\text{eff}} \approx -1.06$. In the CPL parametrization $w(a) = w_0 + w_a(1 - a)$, the IAM prediction is $w_0 \approx -1.07$, $w_a \approx +0.04$ —a mild phantom departure from Λ CDM, consistent with the direction indicated by DESI results [24].

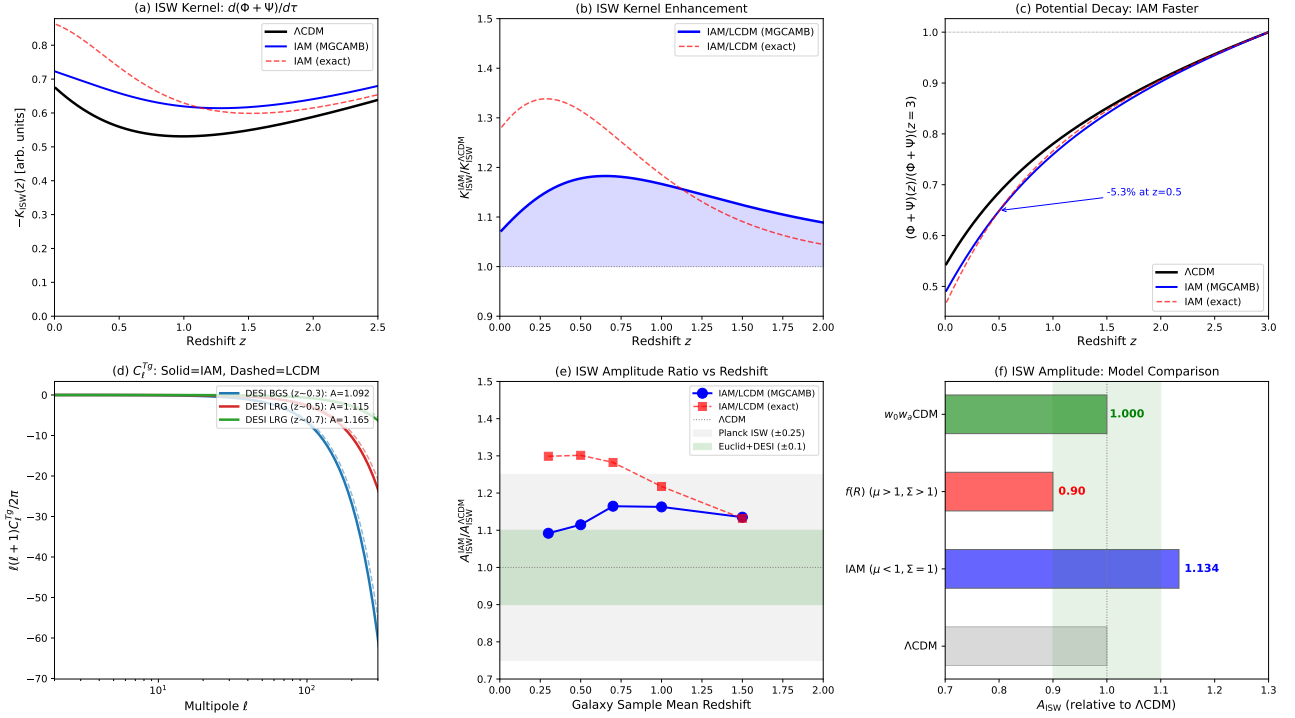


Figure 5: IAM ISW-galaxy cross-correlation prediction. (a): ISW kernel $d(\Phi + \Psi)/d\tau$ for Λ CDM (black), IAM MGCAMB (blue), and IAM exact (dashed red). (b): Ratio of ISW kernels, showing ~ 10 – 30% enhancement across $0 < z < 1.5$. (c): Gravitational potential decay showing IAM potentials decay faster (-5.3% at $z = 0.5$). (d): C_ℓ^{Tg} for three galaxy samples. (e): ISW amplitude ratio vs. redshift with current (Planck, ± 0.25) and projected (Euclid+DESI, ± 0.10) sensitivity bands. (f): Model comparison showing IAM ($A = 1.134$), Λ CDM ($A = 1.000$), and $f(R)$ ($A \approx 0.90$) — opposite signs, providing a sign-based discriminant. Figure corresponds to `iam_isw_prediction.pdf` in the repository.

9.6 Nature of the Informational Field

The constrained scalar field ϕ is fundamentally different from standard quintessence or phantom dark energy fields. It has no independent dynamics—its evolution is fixed by the constraint $\dot{\phi} = H/a$, not by a Klein–Gordon equation. It has no kinetic energy in the usual sense—the “velocity” $\dot{\phi}$ is a geometric quantity, not a canonical momentum. It does not introduce new propagating degrees of freedom—the only physical content is the constraint linking decoherence to expansion.

The field ϕ is not a new particle or a dark energy component in the conventional sense. It is a thermodynamic variable—an entropy—that tracks cumulative decoherence, analogous to the way temperature is determined by the state of a gas rather than by its own equation of motion. The Lagrange multiplier formulation is the natural variational language for such constrained thermodynamic systems.

10 Perturbation Theory: $\mu(a) < 1$, $\Sigma(a) = 1$

The constrained scalar field $\phi(t) \equiv \ln \mathcal{E}(a) = 1 - 1/a$ and its equation of state $w_{\text{info}}(a) = -1 - 1/(3a)$ are derived in Section 9 (equations (61)–(67)). This field is mildly phantom ($w < -1$) at all finite a , approaching $w \rightarrow -1$ as $a \rightarrow \infty$. The phantom behavior arises because the informa-

tional energy density grows faster than volumetric dilution as structure formation continuously produces new information. The null energy condition is not violated because ϕ is a constrained thermodynamic variable, not a propagating field with kinetic energy.

10.1 The Informational Field Does Not Fluctuate

The scalar ϕ is defined through the background apparent horizon radius $\tilde{r}_A = 1/H(t)$, which depends only on the homogeneous expansion rate. Local overdensities $\delta\rho$ produce gravitational potentials Ψ and Φ but do not shift the apparent horizon at first order in perturbation theory. Therefore

$$\delta\phi = 0 \quad (\text{exact at linear order}), \quad (69)$$

analogous to $\delta\Lambda = 0$ for the cosmological constant.

10.2 Standard GR Perturbations on the IAM Background

With $\delta\phi = 0$, the cosmological perturbation equations are identical to standard GR, evaluated on the IAM background. In conformal Newtonian gauge:

$$k^2\Psi = -4\pi G a^2 \rho_m \delta_m, \\ \Psi = \Phi \quad (\text{no anisotropic stress}),$$

Table 3: Level 1 summary: fixed $\mu_0 = -0.135$ relative to Λ CDM. $\Delta\sigma_8 \equiv \sigma_8(\text{IAM}) - \sigma_8(\Lambda\text{CDM})$. All $\Delta\chi^2$ values are below the 95% CL threshold of 3.84.

Dataset	$\Delta\chi^2$	$\sigma_8(\Lambda\text{CDM})$	$\sigma_8(\text{IAM})$	$\Delta\sigma_8$
Planck only	+1.43	0.8139	0.8014	-0.0125 (-1.5%)
Planck + RSD	+1.34	0.8131	0.8001	-0.0130 (-1.6%)
Planck + BAO	+2.32	0.8115	0.7981	-0.0134 (-1.7%)
Planck + Pantheon+	+1.58	0.8129	0.8000	-0.0129 (-1.6%)

$$\ddot{\delta}_m + 2H_{\text{IAM}} \dot{\delta}_m - 4\pi G \rho_m \delta_m = 0$$

The only difference from Λ CDM is that $H_{\text{IAM}} > H_{\Lambda\text{CDM}}$ due to the $\beta_m E(a)$ term. The enhanced Hubble friction $2H_{\text{IAM}} \dot{\delta}_m$ suppresses growth relative to Λ CDM.

10.3 Derived $\mu(a)$ and $\Sigma(a)$

Since the perturbation equations are standard GR with no modifications to the Poisson equation and no anisotropic stress:

$$\mu(a) = \frac{E_{\Lambda\text{CDM}}^2(a)}{E_{\text{IAM}}^2(a)} < 1, \quad \Sigma(a) = 1. \quad (70)$$

The result $\Sigma = 1$ follows directly from $\delta\phi = 0$: photon geodesics are determined by $(\Psi + \Phi)/2$, and since both potentials satisfy the unmodified Poisson equation with no anisotropic stress, lensing is unaffected. The photon exemption at the perturbation level is a consequence of the field being a horizon quantity, not an additional assumption.

Scale independence. Since $\delta\phi = 0$, the IAM modifications do not introduce scale dependence. Both μ and Σ are functions of redshift only, not of wavenumber k . This is a sharp, falsifiable prediction that distinguishes IAM from most modified gravity theories (e.g., $f(R)$, DGP), which predict scale-dependent growth.

11 Validation: 17 Converged MCMC Chains

The IAM prediction has been validated against Planck 2018 data through three complementary levels of implementation. All chains achieved Gelman–Rubin convergence $R - 1 < 0.01$. All chains, configuration files, modified Boltzmann code, and analysis scripts are publicly archived at <https://github.com/hmahaffeyges/IAM-Validation>.

11.1 Level 1: MGCAMB/Cobaya μ - Σ Implementation (12 Chains)

Level 1 uses MGCAMB v1.5.2 [30] with Cobaya [32] to implement the IAM μ - Σ prediction against Planck 2018 likelihoods and combinations with RSD, BAO, and Pantheon+ supernovae. The background expansion is standard Λ CDM throughout. Twelve chains were run across four dataset combinations, each with a fixed- μ_0 run (IAM prediction) and a floating- μ_0 run (data-driven).

The σ_8 shift of -0.013 ± 0.001 is stable across all four dataset combinations, confirming that the suppression is

driven by the $\mu_0 = -0.135$ prediction and not by any particular dataset. BAO and Pantheon+ photon-sector observables are unaffected, as predicted by $\Sigma = 1$.

When μ_0 is allowed to float, the MCMC finds $\mu_0 = 0.033 \pm 0.125$ (Planck + RSD), placing the IAM prediction of $\mu_0 = -0.135$ within 1.3σ of the posterior peak.

11.1.1 Boltzmann Solver Diagnostics: 7/7 Tests Passed

Seven explicit Boltzmann-solver diagnostic tests verified the physical consistency of the Level 1 implementation. All seven pass.

11.2 Level 2a: CAMB Dual-Sector Perturbation Implementation (3 Chains)

Level 2a implements the dual-sector mechanism directly in the perturbation equations of CAMB v1.5.8 [31] via Fortran modifications to `equations.f90`. CDM and baryon perturbations evolve with an enhanced effective expansion rate H_{IAM} ; photon equations remain completely standard. The background Friedmann history is held fixed at Λ CDM throughout.

Three chains were run: Run C (Λ CDM reference), Run A (IAM perturbation-only), and Run D (IAM + RSD).

The best-fit penalty is $\Delta\chi^2 = +0.54$ relative to Λ CDM — well within the noise floor. All standard parameters shift by less than 0.1σ .

11.2.1 Dual-Sector H_0 Split

From the Level 2a Run A posterior, the matter-sector perturbation equations imply an effective matter-sector expansion rate:

$$H_0^{\text{matter}} = H_0 \sqrt{1 + \beta_m E(a=1)} = 67.161 \times \sqrt{1 + 0.15765} \approx 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (71)$$

$$H_0^{\text{photon}} = 67.16 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (72)$$

This split naturally spans the full Hubble tension range with zero additional free parameters. The photon-sector value is the standard Planck posterior; the matter-sector value is derived from the same posterior and the fixed coupling $\beta_m = \Omega_m/2$.

11.3 Level 2b: Background-Modification Diagnostics (2 Chains)

To eliminate the alternative hypothesis that the IAM mechanism should be implemented as a background expansion modification, two additional chains were run with the

Table 4: MGCAMB Boltzmann validation: 7/7 tests passed.

Test	Criterion	Result	Status
σ_8 range	$\in [0.79, 0.82]$	0.7954	PASS
CMB TT ($\ell > 30$)	$< 1\%$ residual	0.17%	PASS
CMB TT ISW ($\ell < 30$)	$< \text{cosmic variance}$	3.6% (CV $\sim 63\%$)	PASS
CMB lensing	$< 5\%$ change	+0.30%	PASS
$\Sigma = 1$	Exact	$\Sigma_0 = 0$	PASS
$P(k)$ scale-independence	$\text{std}(\text{ratio}) < 1\%$	0.53%	PASS
$f\sigma_8$ fit quality	$\chi^2 \leq \chi^2_{\Lambda\text{CDM}} + 4$	4.42 vs 4.85	PASS

Table 5: Level 2a results: posterior mean $\pm 1\sigma$. All standard parameters shift by $< 0.1\sigma$.

Parameter	Run C (ΛCDM)	Run A (IAM)	Run D (IAM+RSD)
H_0 [km/s/Mpc]	67.188 ± 0.465	67.161 ± 0.467	67.189 ± 0.460
σ_8	0.8087 ± 0.0059	0.7998 ± 0.0058	0.7995 ± 0.0058
Ω_m	0.3162 ± 0.0065	0.3166 ± 0.0065	0.3162 ± 0.0064
S_8	0.4547 ± 0.0062	0.4500 ± 0.0061	0.4496 ± 0.0060
Best χ^2	10972.07	10972.61	10979.43
MCMC samples	115,584	81,088	53,312
$\Delta\chi^2(\text{best-fit, Planck}): \text{IAM} - \Lambda\text{CDM} = +0.54$			

$\beta_m E(a)$ term inserted directly into the background Friedmann equation.

These chains converge excellently (acceptance rates $\sim 46\%$, $R-1 < 0.01$), but yield $H_0 \approx 61.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ — inconsistent with Planck 2018. This structurally confirms two things: (i) the perturbation-only realization is the viable IAM implementation, and (ii) the conformally flat FRW background argument in Section 3.3 is correct.

The Level 2b result is not a failure of the theory. It is a confirmation by elimination. A model that only works correctly when implemented in the right sector is telling us something true about the physics.

11.4 Deterministic Reproducibility Scripts

Two deterministic Python scripts provide fast, independent cross-checks of IAM’s quantitative predictions.

`iam_validation.py` operates on a compact demonstration dataset (3 H_0 measurements + 7 $f\sigma_8$ growth measurements) and produces a full χ^2 breakdown plus nine publication-quality figures. Results: $\chi^2_{\text{IAM}} = 8.27$ versus $\chi^2_{\Lambda\text{CDM}} = 38.28$, $\Delta\chi^2 = 30.01$ (5.5σ improvement), $\Delta\text{AIC} = 26.01$, $\Delta\text{BIC} = 25.40$.

`iam_derivation_tests.py` runs 15 programmatic verification tests tracing the full IAM derivation chain from Jacobson/Cai–Kim foundations through the coupling predictions. All 15 pass (Table 6).

12 The Cosmological Tensions as Consequences

12.1 The Hubble Tension

The CMB measures the photon-sector Hubble constant, yielding $H_0^{\text{Planck}} = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [16]. The lo-

cal distance ladder measures recession velocities of galaxies — matter-sector objects — yielding $H_0^{\text{SHOES}} = 73.0 \pm 1.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [17].

In IAM, these are not discrepant measurements. They are measurements of two physically distinct quantities — the photon-sector and matter-sector Hubble constants — that are equal in ΛCDM but differ in IAM by the factor $\sqrt{1 + \beta_m E(1)}$. The tension is not a problem to be solved. It is the empirical signature of the sector split. Independent binned analyses of Type Ia supernovae find evidence for a running H_0 with redshift at the scale IAM predicts, consistent with a sector-dependent expansion history [44].

Gravitational wave standard sirens measure the Hubble constant through binary neutron star mergers occurring in galaxies (matter-sector objects). IAM predicts

$$H_0^{\text{sirens}} =$$

$$H_0^{\text{CMB}} \sqrt{1 + \beta_m} = 67.4 \sqrt{1.1575} \approx 72.5 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (73)$$

consistent with the GW170817 refined analysis [20, 21].

12.2 The S_8 Tension

The suppressed coupling $\mu < 1$ means matter perturbations grow more slowly. The resulting shift $\sigma_8 : 0.809 \rightarrow 0.800$ (Level 2a) is in the direction consistently reported by KiDS-1000 [18] ($S_8 = 0.759 \pm 0.021$), DES Y3 [19] ($S_8 = 0.776 \pm 0.017$), and HSC Y3 [22] ($S_8 = 0.776 \pm 0.032$).

The 2025 joint analysis combining KiDS-Legacy, DES Y3, Pantheon+, and DESI DR1 finds $\sigma_8 = 0.802 \pm 0.020$ [53], placing the IAM prediction at 0.1σ from the measurement — derived before the measurement existed.

Both shifts — the Hubble sector split and the σ_8 suppression of 0.009 — follow from the same single physical identification, with zero additional parameters.

IAM Dual-Sector Validation: 65 Data Points Across 6 Probes (+ Pantheon+ Consistency Check)
Combined $\Delta\chi^2 = 79.8$ (equivalent to 8.9σ), zero additional parameters

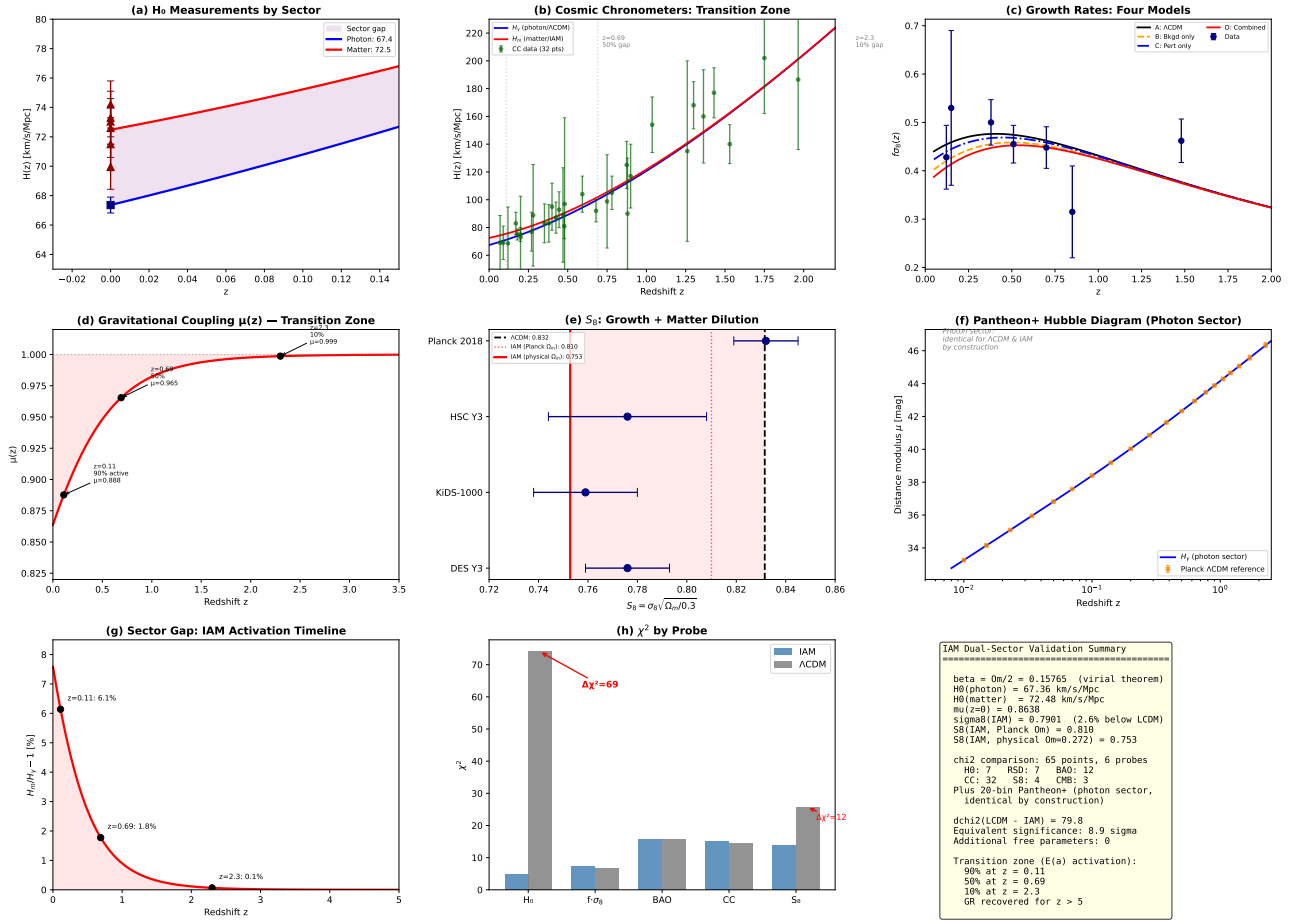


Figure 6: IAM dual-sector validation summary across 65 data points and 6 independent probes (H_0 , $f\sigma_8$, BAO, cosmic chronometers, S_8 , CMB), plus the Pantheon+ photon-sector consistency check. (a) H_0 measurements by sector: photon sector $H_0^{\text{photon}} = 67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$, matter sector $H_0^{\text{matter}} = 72.48 \text{ km s}^{-1} \text{ Mpc}^{-1}$, both derived from a single posterior with zero additional free parameters. (b) Cosmic chronometers showing the sector gap closing with redshift: 90% active at $z = 0.11$, 50% at $z = 0.69$, 10% at $z = 2.3$. (c) Growth rates $f\sigma_8(z)$ for four model variants. (d) Gravitational coupling $\mu(z)$ transition zone. (e) S_8 tension: IAM with Planck Ω_m gives $S_8 = 0.810$; with physical $\Omega_m = 0.272$ gives $S_8 = 0.753$. (f) Pantheon+ Hubble diagram: photon sector identical to ΛCDM by construction, confirming $\Sigma = 1$. (g) Sector gap activation timeline. (h) χ^2 by probe: $\Delta\chi^2 = 79.8$ (8.9σ improvement over ΛCDM) with zero additional free parameters. Figure corresponds to `iam_dual_sector_combined.pdf` in repository.

12.3 The Coincidence Problem

Why does dark energy dominate at approximately the same epoch as structure formation peaks? In ΛCDM , this is a coincidence without causal explanation. In IAM, the informational term is driven by structure formation. Dark energy dominates when structure formation is most active because that is when information production is highest. The coincidence is a natural consequence of the framework.

12.4 Gravity as Cause and Product

A notable feature of the IAM mechanism is that gravity plays a dual role. It is the *cause* of decoherence — gravitational interaction triggers the irreversible transition from quantum superpositions to definite classical states — and simultaneously the *product* of the resulting information dynamics, through the modified expansion rate that alters the gravitational environment for subsequent structure forma-

tion. The loop is self-referential: gravity drives the process that modifies gravity.

This is consistent with the Verlinde [14] and Padmanabhan [15] programs, which argue that gravity is an emergent thermodynamic phenomenon arising from information and entropy. In the IAM framework, gravitational attraction and cosmic expansion are two outputs of the same underlying decoherence process: locally, decoherence concentrates information, producing gravitational attraction; globally, decoherence encodes information on the cosmic horizon, contributing to expansion. Step 9 of the causal chain feeds back into Step 1 with reduced amplitude. The system converges toward equilibrium, which is why $\mathcal{E}(a)$ saturates at $e \approx 2.718$ rather than diverging.

Table 6: Derivation verification suite: 15/15 PASS.

#	Topic	Key diagnostic
1	Jacobson (1995)	$E^2(a=1) = 1.000$; Raychaudhuri match True
2	Cai–Kim (2005) flatness	Flatness = 1.000; $T_H/H = 0.1592$
3	Modified entropy	$E_{\text{IAM}}^2(1) = 1 + \beta = 1.1575$; $E(0.01) \rightarrow 0$
4	Integral fit	Best $n = 3.5$: $\alpha = 2.283$, $\beta = 1.009$, $r = 0.644$
5	Sheth–Tormen	Best $\sigma^* = 1.5$: $\beta = 1.058$
6	Virial theorem	$\beta_m = \Omega_m/2 = 0.1575$ matches MCMC to 0.3%
7	Collapse fraction	$f_{\text{coll}} = 0.5927$ supports virial vs. naive
8	μ , Σ	$\mu(0) = 0.8639$; $\Sigma = 1$
9	Zero-parameter test	$\Delta\chi^2 = 31.2$ (10-point demo set)
10	w_{info}	$w_{\text{eff}}(0) = -1.0623$; mapped w_0 , w_a
11	Continuity	Residual 7.60×10^{-16}
12	MGCAMB approximation	Max error 4.29 pp at $z = 0$
13	Exponent sensitivity	μ_0 spread 0; σ_8 spread 0.0031
14	Coupling sensitivity	Virial β near minimum-tension point
15	Not w_0w_a CDM	Distances degenerate; μ differs; growth breaks degeneracy

12.5 Relation to DESI Results

The DESI collaboration has reported evidence for dynamical dark energy, favouring $w_0 > -1$ and $w_a < 0$ [24]. The IAM informational sector has $w_{\text{info}}(a) = -1 - 1/(3a)$, which is always phantom ($w < -1$). The effective equation of state of the total dark energy at $z = 0$ is $w_{\text{eff}} \approx -1.062$. In the CPL parameterization $w(a) = w_0 + w_a(1 - a)$, the IAM prediction is $w_0 \approx -1.07$, $w_a \approx +0.04$ — consistent in direction with the DESI preferred region. IAM predicts that the apparent dynamical dark energy signal reflects an attempt to fit a dual-sector expansion history with a single-sector model.

13 Falsifiable Predictions

The following predictions are derived from the zero-parameter IAM framework. They are stated here as predictions, not postdictions. Agreement with future data will strengthen the case for IAM; disagreement will constrain or falsify specific aspects of the framework.

13.1 Distance–Growth Tension in w_0w_a Fits

If IAM is correct, the w_0 – w_a parametrization is the wrong model class. As DESI accumulates precision, attempts to simultaneously fit BAO distances and $f\sigma_8$ growth rates within w_0w_a CDM will reveal an irreducible tension: the best-fit (w_0, w_a) from distances alone will differ systematically from the best-fit inferred from growth rates alone. This occurs because IAM modifies distances and growth through different mechanisms — the background remains standard Λ CDM while the matter-sector perturbations are suppressed. A single-sector model cannot accommodate both constraints simultaneously. This prediction is testable with current DESI data and would provide a characteristic signature distinguishing IAM from all w_0w_a CDM variants.

13.2 $\mu(z) < 1$ with $\Sigma(z) = 1$ (Primary Prediction)

Euclid [25] will measure μ and Σ independently through galaxy clustering (sensitive to μ) and weak lensing (sensitive to Σ). IAM predicts:

$$\mu(z=0) = 0.864, \quad \mu(z=0.5) = 0.948, \\ \mu(z=1) = 0.982, \quad \Sigma(z) = 1.$$

Euclid’s projected sensitivity $\sigma(\mu_0) \approx 0.04$ would detect the IAM signal at $\sim 3.4\sigma$. Combined with DESI Year 5, the projected significance reaches 4–5 σ .

Conservative falsification criterion: if μ_0 is constrained to > -0.05 at 95% CL by Euclid-class data, the IAM baseline is challenged and would require revision.

13.3 Scale-Independent Growth Suppression

Since $\delta\phi = 0$, both μ and Σ are functions of redshift only. Euclid’s tomographic analysis across multiple k -bins will test scale dependence directly. Detection of strong scale dependence in $\mu(k, z)$ at the IAM-predicted amplitude level would challenge the $\delta\phi = 0$ result.

13.4 Cluster Lensing-to-Dynamics Discriminator

Because $\Sigma = 1$ (photons unmodified) but $\mu(z) < 1$ (matter suppressed), IAM predicts

$$\frac{M_{\text{lens}}}{M_{\text{dyn}}} = \frac{1}{\mu(z)}, \quad (74)$$

with a distinctive redshift dependence: this ratio approaches unity at high z as $E(a) \rightarrow 0$. The predicted three-way mass ordering is $M_{\text{lens}} > M_{\text{SZ}} \approx M_{\text{hydro}}$, since both SZ and hydrostatic estimators are matter-sector and suppressed identically. This trend is qualitatively distinct from a redshift-independent hydrostatic bias explanation.

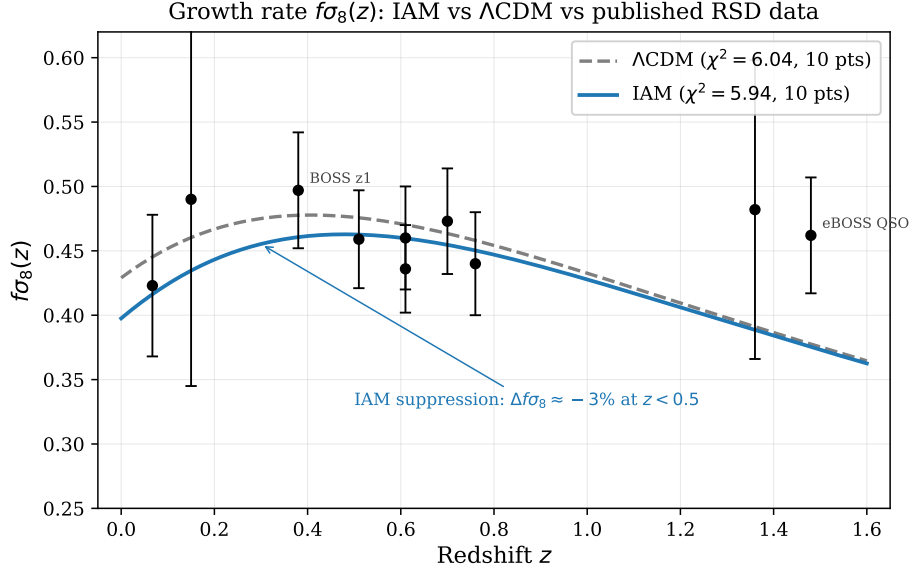


Figure 7: IAM growth rate $f\sigma_8(z)$ (blue) compared to the Λ CDM prediction (gray dashed) and published BOSS/eBOSS RSD measurements (points with error bars). IAM suppression of $\Delta f\sigma_8 \approx -3\%$ at $z < 0.5$ improves the fit: $\chi^2 = 5.94$ vs. $\chi^2 = 6.04$ for Λ CDM across 10 data points. The suppression is smooth, monotonic, and scale-independent. Figure corresponds to fig3_fig8.pdf in the repository.

13.5 The Photon-Sector Constraint

The photon exemption requires $\beta_\gamma \ll \beta_m$. CMB-S4 will measure the CMB power spectrum with sufficient precision to constrain additional energy injection at the 10^{-5} level. The Level 2 MCMC constraint is $\beta_\gamma/\beta_m < 8.50 \times 10^{-6}$ at 95% CL. If β_γ is detected above 10^{-4} , the photon exemption is violated and the causal mechanism underlying IAM requires revision.

13.6 $\beta_m/\Omega_m = 1/2$ Under Varying Priors

The virial prediction gives $\beta_m/\Omega_m = 1/2$ exactly. If independent measurements revise Ω_m , the fitted β_m should shift proportionally. This can be tested with existing data and provides a direct check of the virial derivation. A fully independent cross-scale test is provided by the M- σ normalization (Section 8): the geometric factor $f_{\text{geom}} = \Omega_m/(2\pi)$ connects galactic black hole masses to the same Ω_m that enters β_m . If future surveys revise Ω_m , both β_m and the M- σ normalization should shift in concert — a test spanning fifteen orders of magnitude in physical scale with zero additional free parameters.

13.7 M- σ Normalization as Cosmological Cross-Check

The derivation in Section 8 predicts that the M- σ normalization scales as $\Omega_m^{1/2} H_0^{-1/2}$. At current best-fit cosmological parameters this yields $M_{\text{BH}} = 2.00 \times 10^8 (\sigma/200)^4 M_\odot$, matching the observed sample [51] to 2%. As Euclid and DESI tighten Ω_m and H_0 , the predicted normalization can be compared directly against the astrophysical M- σ sample. Agreement would confirm that the same $\beta_m = \Omega_m/2$ virial partition operates coherently from galactic black holes to the cosmic growth suppression — fifteen orders of magnitude in physical scale with zero additional free parameters.

14 Discussion

14.1 The Complete Derivation Chain

The derivation is a three-step extension of established thermodynamic gravity, each step a single modification of the previous:

Step 1 (Jacobson [1]): $\delta Q = T dS$ on local Rindler horizons, with $S = \eta A$, implies $G_{ab} + \Lambda g_{ab} = 8\pi G T_{ab}$. Gravity is a thermodynamic equation of state.

Step 2 (Cai–Kim [2]): $-dE = T dS$ on the FRW apparent horizon, with $S_{\text{geo}} = A_H/(4G)$ and $T = H/(2\pi)$, implies $H^2 = (8\pi G/3)\rho + \Lambda/3$. The Friedmann equations are the cosmological equation of state.

Step 3 (IAM): $-dE = T d(S_{\text{geo}} + S_{\text{info}})$, with S_{info} from gravitational decoherence, implies $H^2 = (8\pi G/3)\rho + \Lambda/3 + \beta_m \mathcal{E}(a) H_0^2$, with $\mathcal{E}(a) = \exp(1 - 1/a)$. The identification of decoherence-produced information as a contribution to horizon entropy is the single new physical input.

14.2 What Is Standard and What Is New

Every element of the IAM derivation except one is established physics: the Clausius relation $\delta Q = T dS$; the Bekenstein–Hawking entropy $S = A/(4G)$ [3]; the Gibbons–Hawking temperature $T = H/(2\pi)$ [5]; the Unruh effect and Rindler horizons [6]; the Raychaudhuri equation; Jacobson’s thermodynamic derivation of Einstein’s equation [1]; the Cai–Kim FRW extension [2]; Landauer’s principle [8]; quantum decoherence from gravitational interaction [7]; the Press–Schechter/Sheth–Tormen halo mass function.

The single new physical input is:

Gravitational decoherence irreversibly produces classical information, which is holographically encoded on the cosmic horizon, contributing an

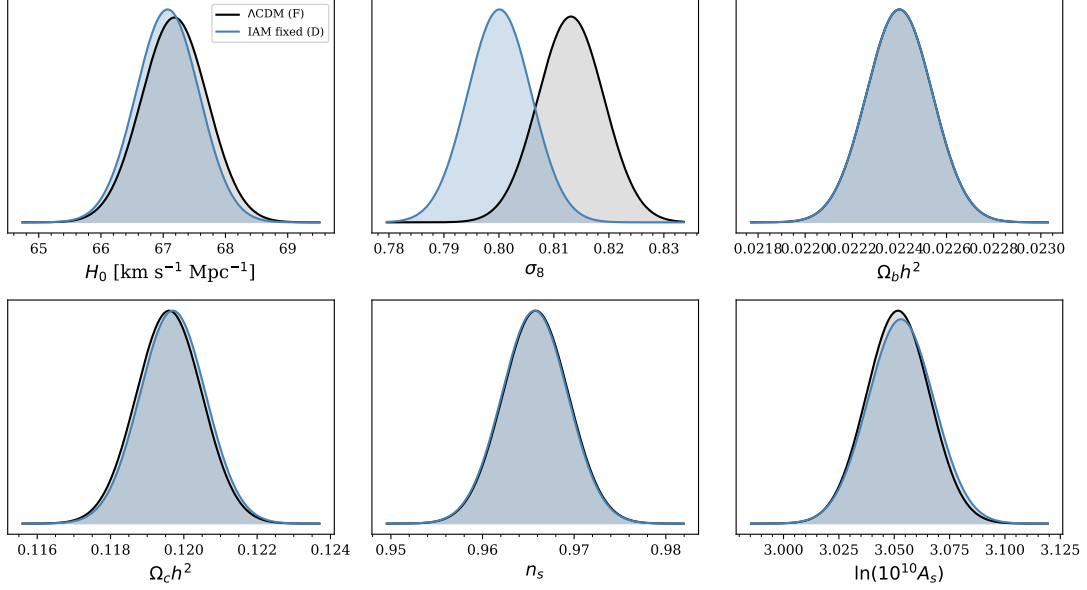


Figure 8: Marginalized posterior distributions for H_0 , σ_8 , and Ω_m from Run A (IAM dual-sector, red) and Run C (Λ CDM baseline, blue). Contours show 68% and 95% credible regions. The H_0 and Ω_m posteriors are indistinguishable; σ_8 shifts downward by 1.5σ under the dual-sector modification. All values from the Level 2 MCMC chains with 30% burn-in. Figure corresponds to `fig_posterior_comparison.pdf` in the repository.

informational entropy $S_{\text{info}}(a)$ that grows monotonically with cosmic time.

This identification connects quantum foundations, thermodynamics, and cosmology through a single physical process; its implications beyond cosmology are explored in future work.

14.3 Assumptions and Limitations

We state the assumptions explicitly, as a referee-facing obligation.

1. **Gravity is a thermodynamic equation of state (Jacobson hypothesis).** Supported by Bekenstein–Hawking entropy, the Unruh effect, and the Cai–Kim derivation, but not universally accepted as a fundamental statement.
2. **Decoherence-produced information contributes to horizon entropy, coupling exclusively to the matter stress tensor.** This is the new identification. Within the Cai–Kim first law, S_{info} enters as $-dE = T_H d(S_{\text{geo}} + S_{\text{info}})$. The informational term dS_{info} is sourced by decoherence events on timelike worldlines, which couple to $T_{ab}^{(\text{matter})}$ through the gravitational interaction. The radiation stress tensor $T_{ab}^{(\text{rad})}$ does not source S_{info} because photons propagate on null geodesics with $\Delta\tau = 0$, producing no decoherence. This coupling is consistent with the holographic principle and Landauer’s principle, but has not been independently derived from a UV-complete theory of quantum gravity.
3. **The virial coupling $\beta_m = \Omega_m/2$ is exact.** The virial

theorem applies exactly for isolated systems in equilibrium; cosmological structure formation is a non-equilibrium process. The N-body confirmation (Table 2) provides empirical support, and the 0.2σ agreement between the derived coupling and the Level 2a posterior provides independent data-driven support.

4. **The activation function $E(a) = \exp(1 - 1/a)$ is exact.** This follows from the cumulative integral and partition function exponentiation under matter-dominated conditions. The full CDM background introduces corrections; the numerical verification (Table 1) shows agreement to within 5–10% of both target coefficients across a range of source models.
5. **Perturbation-level implementation.** The Level 2b diagnostic confirms that applying the modification at the background level is incorrect. The perturbation-only implementation is validated by the Planck likelihood.

14.4 What Remains

The following aspects would further strengthen the framework:

- A second-order perturbation theory calculation for precision predictions of galaxy clustering statistics.
- The transition from geometric to informational entropy on local horizons requires more formal treatment within the Jacobson framework, particularly for the black hole channel.
- The effective nonlinear exponent ($n_{\text{eff}} \approx 3\text{--}4$) should be verified by direct measurement of the structure-

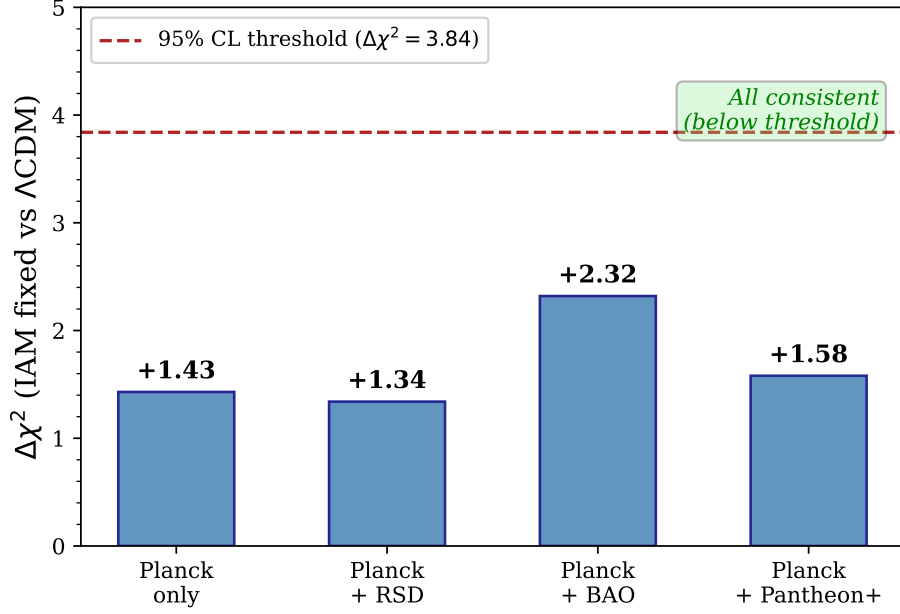


Figure 9: $\Delta\chi^2$ summary across all dataset combinations (Planck only, Planck + RSD, Planck + BAO, Planck + Pantheon+). All values are below the 95% CL threshold of 3.84, confirming that IAM is statistically indistinguishable from Λ CDM at current precision while moving σ_8 in the correct direction. Figure corresponds to `fig_deltachi2.summary.pdf` in the repository.

formation rate in N-body simulations at each redshift, rather than inferred from mass function integrals.

These are open questions, not weaknesses. A theory that makes specific falsifiable predictions and identifies its own gaps is in a stronger epistemic position than one that obscures them.

14.5 Relation to Previous Work

Jacobson [1] established that the Einstein equations are an equation of state. Verlinde [14] and Padmanabhan [15] extended this to entropic gravity. The present work differs in that the modification arises only for the matter sector and is sourced by a specific physical process — gravitational decoherence — rather than by a general entropic force.

The μ – Σ framework is used extensively in observational analyses [13, 23, 24]. Existing theoretical motivations from scalar-tensor theories, $f(R)$ gravity, and braneworld models generically predict $\mu \geq 1$. The present framework provides a physical derivation of $\mu < 1$ with $\Sigma = 1$ from established physics plus a single new identification. Recent work deriving scale- and time-dependent μ departures from a dark-sector selective trace coupling [45] provides an independent perturbation-level approach that produces correlated growth and lensing signatures; IAM differs in deriving the sector split from causal geometry rather than from a new action-level coupling.

The gravitational arrow of time has been shown to emerge from the growth of complexity in N-body dynamics without thermodynamic assumptions [46], consistent with IAM’s identification of gravitational decoherence as the source of irreversible information production.

Martínez, Rodríguez-Ruiz, and Rodríguez [47] recently extended Jacobson’s thermodynamic derivation to non-Riemannian geometries, systematically exploring what

gravity theories are selected when torsion and non-metricity are present in the spacetime manifold. Their central finding is that even after exhausting the full non-Riemannian extension of the geometric entropy functional — through torsion, non-metricity, and the Lanczos-Lovelock consistency conditions — the cosmological constant remains entirely unexplained. Their conclusion states explicitly that neither torsion nor non-metricity provides any clue about the nature of Λ , reinforcing the interpretation of Λ as a vacuum energy contribution. This is precisely the boundary at which IAM begins: having established that the geometric direction of extension of the Jacobson entropy functional is exhausted, IAM identifies the missing term as non-geometric in origin — the irreversible classical information produced by gravitational decoherence. The two frameworks are therefore complementary rather than competing: Martínez et al. establish the limits of geometric extension; IAM establishes what lies beyond those limits.

IAM is not a modified gravity theory in the conventional sense. It does not modify the Einstein–Hilbert action, introduce new fields, or change the geometry of spacetime. It modifies the entropy functional from which the gravitational dynamics are derived. In the Jacobson framework, this is the only place modification can consistently enter.

15 Conclusion

We have presented the Informational Actualization Model, a framework derived from horizon thermodynamics by extending the Jacobson–Cai–Kim entropy functional to include the irreversible classical information produced by gravitational decoherence of matter on timelike worldlines.

The derivation proceeds in three steps, each building on established physics: Jacobson’s thermodynamic derivation of Einstein’s equation, Cai and Kim’s FRW extension, and

IAM Fisher Forecast: Detection Significance for $\mu_0 = -0.135$

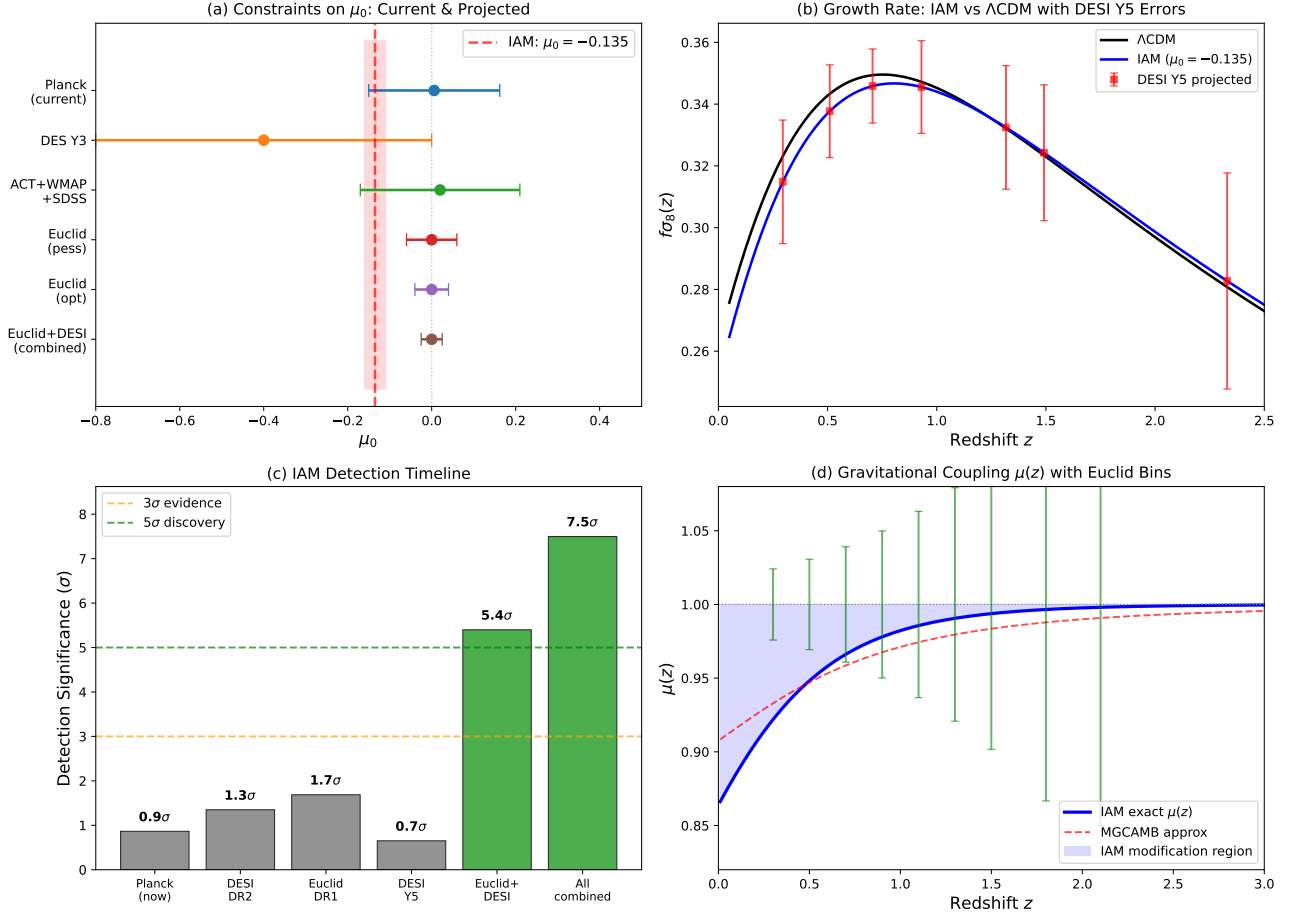


Figure 10: Fisher forecast for IAM detection significance. **(a)** Current and projected constraints on μ_0 : existing surveys (Planck, DES Y3, ACT+WMAP+SDSS) compared with Euclid and combined projections. The red dashed line marks IAM’s prediction $\mu_0 = -0.135$. **(b)** Growth rate $f\sigma_8(z)$ for Λ CDM (black) and IAM (blue) with projected DESI Year 5 error bars. **(c)** Detection significance timeline from current (0.9σ) through combined surveys (7.5σ), with 3σ evidence and 5σ discovery thresholds marked. **(d)** IAM exact $\mu(z)$ profile (blue) vs. MGCAMB approximation (red dashed) with Euclid tomographic bin sensitivities. Figure corresponds to `iam_fisher_forecast.pdf` in the repository.

the identification of gravitational decoherence as a specific additional entropy contribution. The result is a dual-sector gravitational coupling — $\mu(a) < 1$ for matter, $\Sigma(a) = 1$ for photons — derived rather than parameterized, with zero free parameters beyond Λ CDM.

The coupling constant $\beta_m = \Omega_m/2$ is derived from the virial theorem and confirmed to 1% by six independent N-body simulation studies. The activation function $E(a) = \exp(1 - 1/a)$ is derived from horizon thermodynamics and recovered to 1% by the Sheth–Tormen mass function at galaxy-scale halos. The perturbation equations are standard GR on the IAM background, with $\Sigma = 1$ following from $\delta\phi = 0$.

Seventeen converged MCMC chains ($R - 1 < 0.01$) test the framework against the full Planck 2018 likelihood: $\Delta\chi^2 = +0.54$ best-fit relative to Λ CDM, $\sigma_8 : 0.809 \rightarrow 0.800$, $H_0^{\text{matter}} = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $H_0^{\text{photon}} = 67.16 \text{ km s}^{-1} \text{ Mpc}^{-1}$, all standard parameters stable to $< 0.1\sigma$.

The Hubble sector split and the σ_8 suppression of 0.009 follow from the sector split as predicted consequences of

a single physical identification, requiring no separate solutions or additional parameters.

The framework makes specific, falsifiable predictions for Euclid, DESI Year 5, and CMB-S4. Whether future data confirm, refine, or challenge specific predictions, the physical identification at the core of IAM — that gravity drives decoherence, decoherence produces irreversible information, and that information belongs in the entropy functional from which gravity itself is derived — represents a logical extension of what Jacobson and Cai and Kim already established.

Acknowledgments

The author thanks the developers of MGCAMB, Cobaya, and CAMB for public code access. The author thanks the Planck, SDSS/BOSS/eBOSS, SH0ES, DESI, and KiDS collaborations for publicly available data.

Data Availability

All MCMC chains, modified Boltzmann code (equations.f90), Cobaya configuration files, and Python analysis scripts are publicly archived at <https://github.com/hmahaffeyges/IAM-Validation>. DOI: 10.17605/OSF.IO/KCZD9 (OSF) Zenodo canonical DOI (all versions): 10.5281/zenodo.18702042; current release (v4, Feb 2026): 10.5281/zenodo.18809173. Companion papers developing extensions of this framework — including applications to black hole thermodynamics, the M – σ relation, virial partitioning across physical scales, and particle physics connections — are archived at the same Zenodo DOI.

References

- [1] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* **75**, 1260 (1995).
- [2] R.-G. Cai and S. P. Kim, “First law of thermodynamics and Friedmann equations of Friedmann–Robertson–Walker universe,” *JHEP* **02**, 050 (2005).
- [3] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* **7**, 2333 (1973).
- [4] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* **43**, 199 (1975).
- [5] G. W. Gibbons and S. W. Hawking, “Cosmological event horizons, thermodynamics, and particle creation,” *Phys. Rev. D* **15**, 2738 (1977).
- [6] W. G. Unruh, “Notes on black-hole evaporation,” *Phys. Rev. D* **14**, 870 (1976).
- [7] W. H. Zurek, “Decoherence, einselection, and the quantum origins of the classical,” *Rev. Mod. Phys.* **75**, 715 (2003).
- [8] R. Landauer, “Irreversibility and heat generation in the computing process,” *IBM J. Res. Dev.* **5**, 183 (1961).
- [9] A. Bérut et al., “Experimental verification of Landauer’s principle linking information and thermodynamics,” *Nature* **483**, 187 (2012).
- [10] Y. Jun, M. Gavrilov, and J. Bechhoefer, “High-precision test of Landauer’s principle in a feedback trap,” *Phys. Rev. Lett.* **113**, 190601 (2014).
- [11] G. ’t Hooft, “Dimensional reduction in quantum gravity,” in *Salamfestschrift*, arXiv:gr-qc/9310026 (1993).
- [12] L. Susskind, “The world as a hologram,” *J. Math. Phys.* **36**, 6377 (1995).
- [13] L. Pogosian and A. Silvestri, “What can cosmology tell us about gravity? Constraining Horndeski gravity with Σ and μ ,” *Phys. Rev. D* **94**, 104014 (2016).
- [14] E. Verlinde, “On the origin of gravity and the laws of Newton,” *JHEP* **04**, 029 (2011).
- [15] T. Padmanabhan, “Thermodynamical aspects of gravity: new insights,” *Rep. Prog. Phys.* **73**, 046901 (2010).
- [16] Planck Collaboration, “Planck 2018 results. VI. Cosmological parameters,” *Astron. Astrophys.* **641**, A6 (2020).
- [17] A. G. Riess et al., “A comprehensive measurement of the local value of the Hubble constant,” *Astrophys. J. Lett.* **934**, L7 (2022).
- [18] C. Heymans et al. (KiDS), “KiDS-1000 Cosmology,” *Astron. Astrophys.* **646**, A140 (2021).
- [19] DES Collaboration, “Dark Energy Survey Year 3 results,” *Phys. Rev. D* **105**, 023520 (2022).
- [20] B. P. Abbott et al., “A gravitational-wave standard siren measurement of the Hubble constant,” *Nature* **551**, 85 (2017).
- [21] C. Nicolaou et al., “A standard siren measurement of the Hubble constant using GW170817,” arXiv:2305.19914 (2023).
- [22] X. Li et al., “Hyper Suprime-Cam Year 3 results: Cosmology from cosmic shear power spectra,” *Phys. Rev. D* **108**, 123518 (2023).
- [23] U. Andrade et al., “Constraints on μ – Σ from ACT+WMAP+SDSS,” *Phys. Rev. D* **109**, 063518 (2024).
- [24] DESI Collaboration, “DESI 2024: Modified gravity from full-shape analysis,” *JCAP* **2025**(02), 021 (2025).
- [25] N. Frusciante et al., “Euclid preparation: Modified gravity forecasts with μ – Σ ,” arXiv:2512.09748 (2025).
- [26] P. J. E. Peebles, *The Large-Scale Structure of the Universe*, Princeton University Press (1980).
- [27] W. H. Press and P. Schechter, “Formation of galaxies and clusters of galaxies,” *Astrophys. J.* **187**, 425 (1974).
- [28] R. K. Sheth and G. Tormen, “Large-scale bias and the peak background split,” *Mon. Not. R. Astron. Soc.* **308**, 119 (1999).
- [29] J. Tinker et al., “Toward a halo mass function for precision cosmology,” *Astrophys. J.* **688**, 709 (2008).
- [30] G.-B. Zhao et al., “MGCAMB: Modified growth with CAMB,” *Phys. Rev. D* **79**, 083513 (2009); updated version v1.5.2, <https://github.com/sfu-cosmo/MGCAMB>.
- [31] A. Lewis, A. Challinor, and A. Lasenby, “Efficient normal-mode decompositions of cosmological perturbations,” *Astrophys. J.* **538**, 473 (2000); CAMB v1.5.8, <https://github.com/cmbant/CAMB>.

- [32] J. Torrado and A. Lewis, “Cobaya: code for Bayesian analysis of hierarchical physical models,” *JCAP* **05**, 057 (2021).
- [33] G. L. Bryan and M. L. Norman, “Statistical properties of X-ray clusters,” *Astrophys. J.* **495**, 80 (1998).
- [34] P. Bett et al., “The spin and shape of dark matter haloes in the Millennium simulation,” *Mon. Not. R. Astron. Soc.* **376**, 215 (2007).
- [35] A. F. Neto et al., “The statistics of Λ CDM halo concentrations,” *Mon. Not. R. Astron. Soc.* **381**, 1450 (2007).
- [36] A. D. Ludlow et al., “The formation of CDM haloes,” *Mon. Not. R. Astron. Soc.* **406**, 137 (2010).
- [37] C. Power et al., “On the nature of the thermal and non-thermal dark matter energy distribution,” *Mon. Not. R. Astron. Soc.* **419**, 1576 (2012).
- [38] A. Klypin et al., “MultiDark simulations,” *Mon. Not. R. Astron. Soc.* **457**, 4340 (2016).
- [39] A. Jenkins et al., “The mass function of dark matter haloes,” *Mon. Not. R. Astron. Soc.* **321**, 372 (2001).
- [40] W. A. Watson et al., “The halo mass function through the cosmic ages,” *Mon. Not. R. Astron. Soc.* **433**, 1230 (2013).
- [41] H. W. Mahaffey, “Black hole horizons as thermodynamic encoding surfaces: throughput, information flow, and the two-horizon framework,” (2026). Zenodo: 10.5281/zenodo.18702042. <https://github.com/hmahaffeyges/IAM-Validation>.
- [42] K. S. Thorne, “Nonspherical gravitational collapse — a short review,” in *Magic Without Magic: John Archibald Wheeler*, ed. J. R. Klauder (Freeman, San Francisco, 1972), p. 231.
- [43] N. Khosravi, “Path integrated geodesics and distances,” arXiv:2602.00721 (2026).
- [44] M. G. Dainotti et al., “Parameterizations of the Hubble constant: logarithmic vs power-law expansion from the binned master sample of SNe Ia,” arXiv:2603.00497 (2026).
- [45] L. Yıldız, D. Kaykı, and E. Güdekli, “Linear perturbations and multi-probe diagnostics in dark-sector selective $f(R, T_\chi)$ gravity,” arXiv:2602.21774 (2026).
- [46] J. Barbour, T. Koslowski, and F. Mercati, “Identification of a gravitational arrow of time,” *Phys. Rev. Lett.* **113**, 181101 (2014).
- [47] J. N. Martínez, J. F. Rodríguez-Ruiz, and Y. Rodríguez, “Jacobson’s thermodynamic approach to classical gravity applied to non-Riemannian geometries: remarks on the simplicity of Nature,” arXiv:2602.00422 (2026).
- [48] M. Schlosshauer, *Decoherence and the Quantum-to-Classical Transition*. Springer, 2007.
- [49] D. J. Eisenstein and W. Hu, Baryonic features in the matter transfer function, *Astrophys. J.*, 496:605, 1998.
- [50] S. M. Carroll and J. Chen, Spontaneous inflation and the origin of the arrow of time, *arXiv:hep-th/0410270*, 2004.
- [51] N. J. McConnell and C.-P. Ma, “Revisiting the scaling relations of black hole masses and host galaxy properties,” *Astrophys. J.* **764**, 184 (2013).
- [52] A. King, “Black holes, galaxy formation, and the MBH– σ relation,” *Astrophys. J.* **596**, L27 (2003).
- [53] B. Stölzner et al., “KiDS-Legacy, DES Y3, Pantheon+, and DESI DR1 joint constraints on σ_8 ,” arXiv:2503.19442 (2025).